CardamomOT — analysis report

`/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Kameneva`

Generated on 2026-10-19 16:23

Run

Split	train
Stimulus penalisation	0.2
Prior penalisation	1.0
Genes	97
Stimuli	1
Observed cells	1260
Timepoints	0, 8, 32, 48, 64, 80
Cell types	precursors, intermediate, sympathoblasts, chromaffin
Network shown	inter_simul.npy
Embeddings	UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (counts / depth factor, model draws taken at the depth of the cells they mimic)

Perturbations (perturbation_simulation)

- ✓ P1 KO CHGA (50%)
- ✓ P2 KO CHGA
- ✓ P3 OV STMN2
- ✓ P4 KO CHGA · OV STMN2

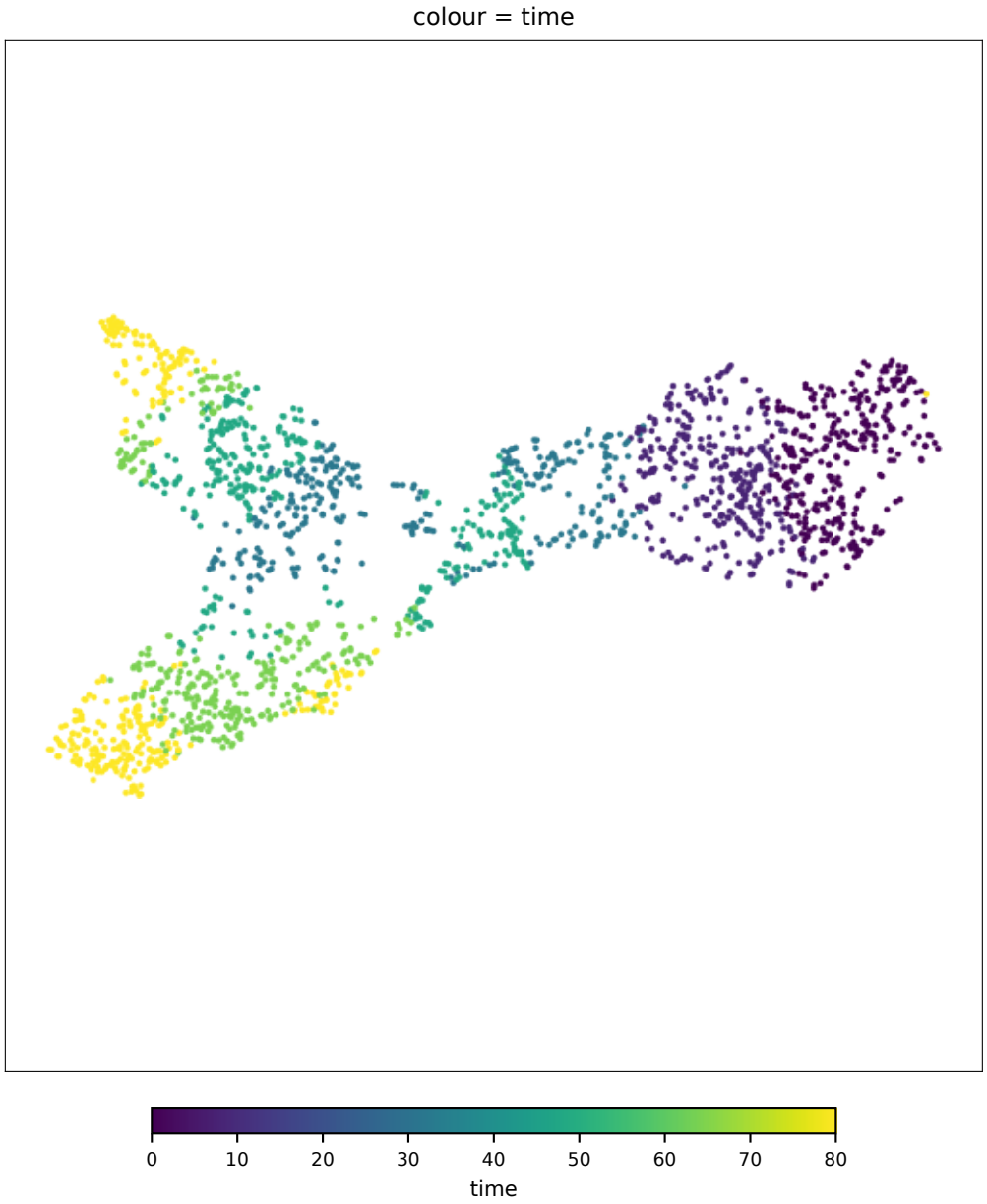
Contents

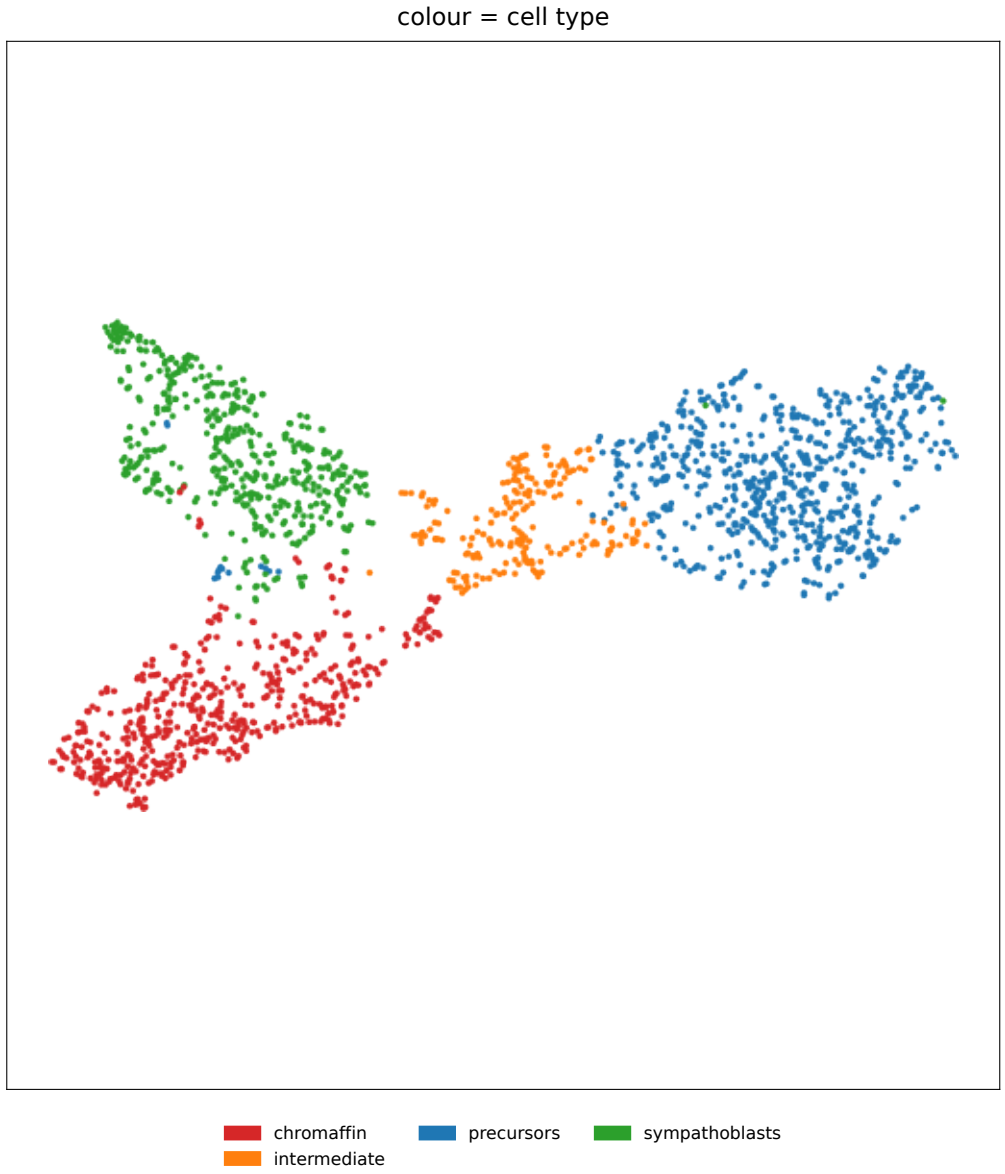
Data — UMAP of all the cells of Data/data.h5ad (time, samples), also on the genes of the model with select_genes; teaser: classical OT vs CardamomOT displacement fields (

1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots), top-20 regulators and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

Data — UMAP of all the cells

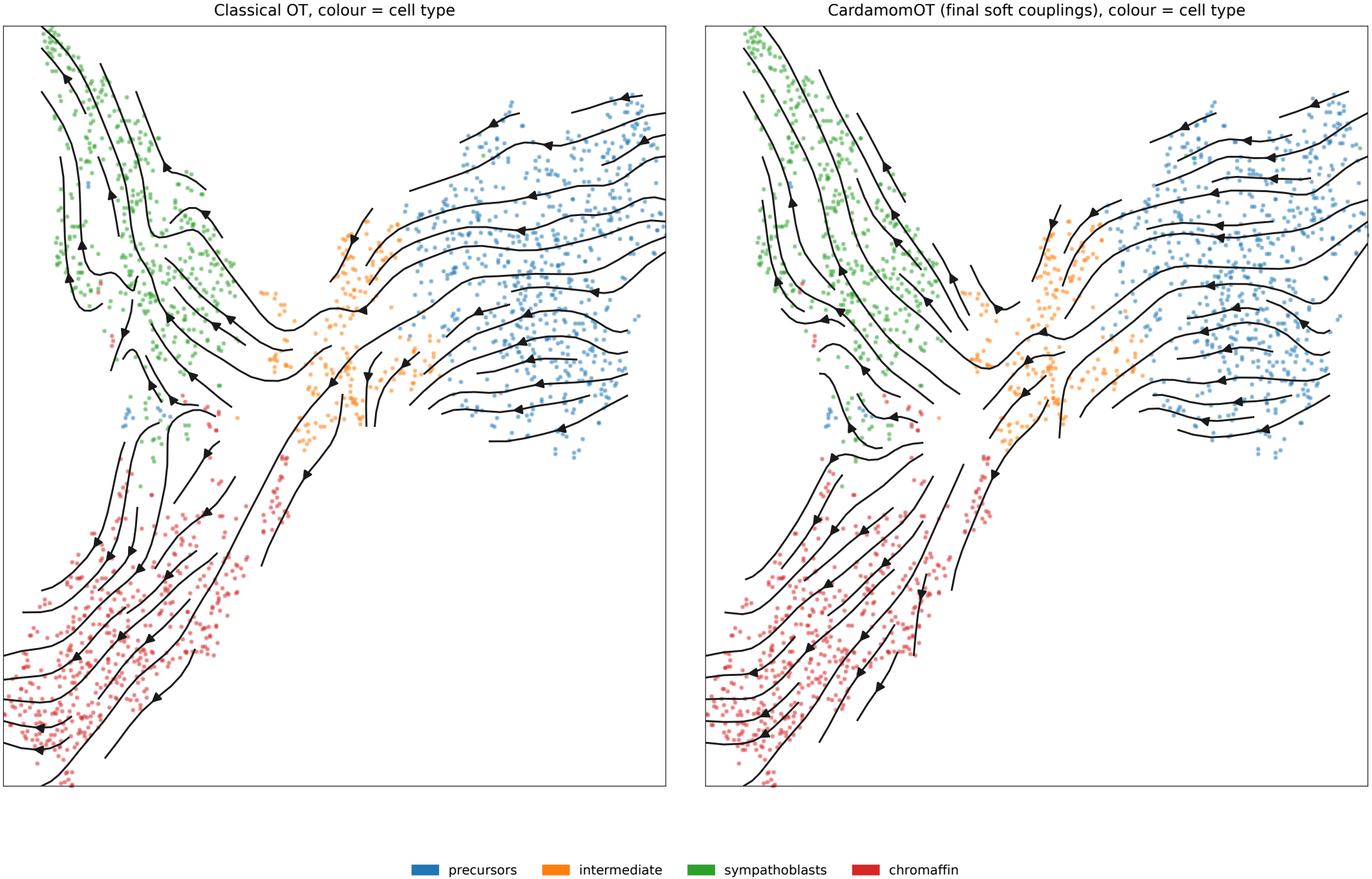
Data/data.h5ad, 1800 cells: log1p(counts), 97 highly variable genes, PCA (50 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data.npz.





Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.68 (PCA space). CardamomOT: final soft couplings of the trajectories (1260 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



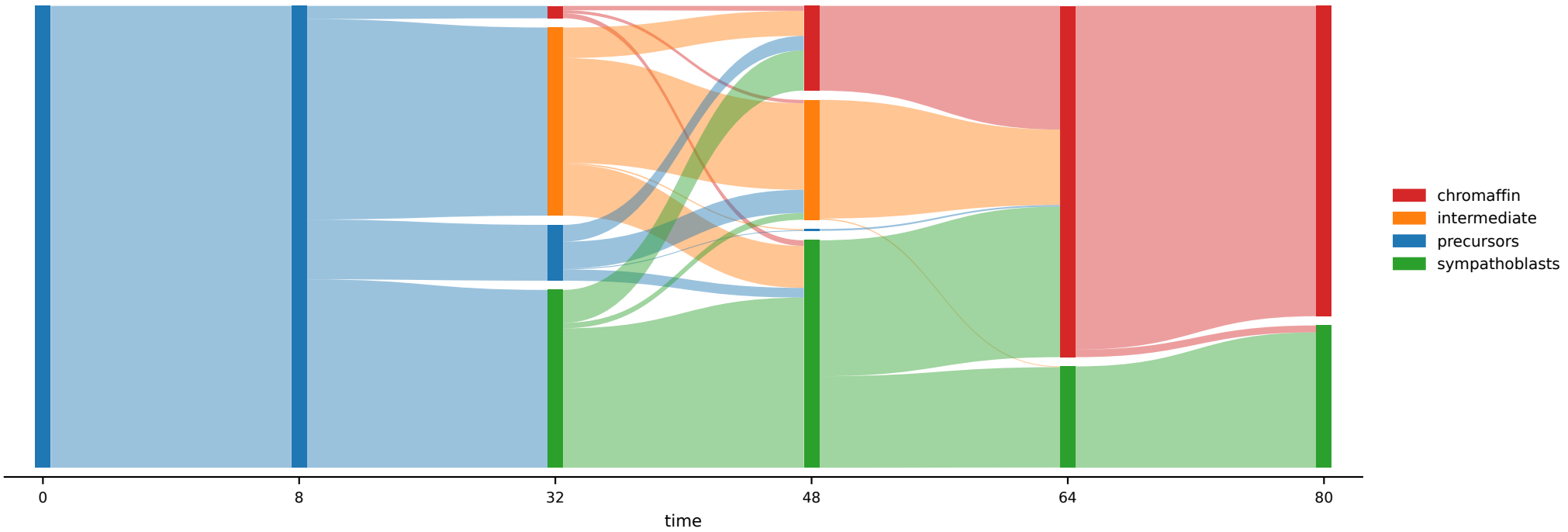
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



Classical OT — cell-type transitions and fate genes

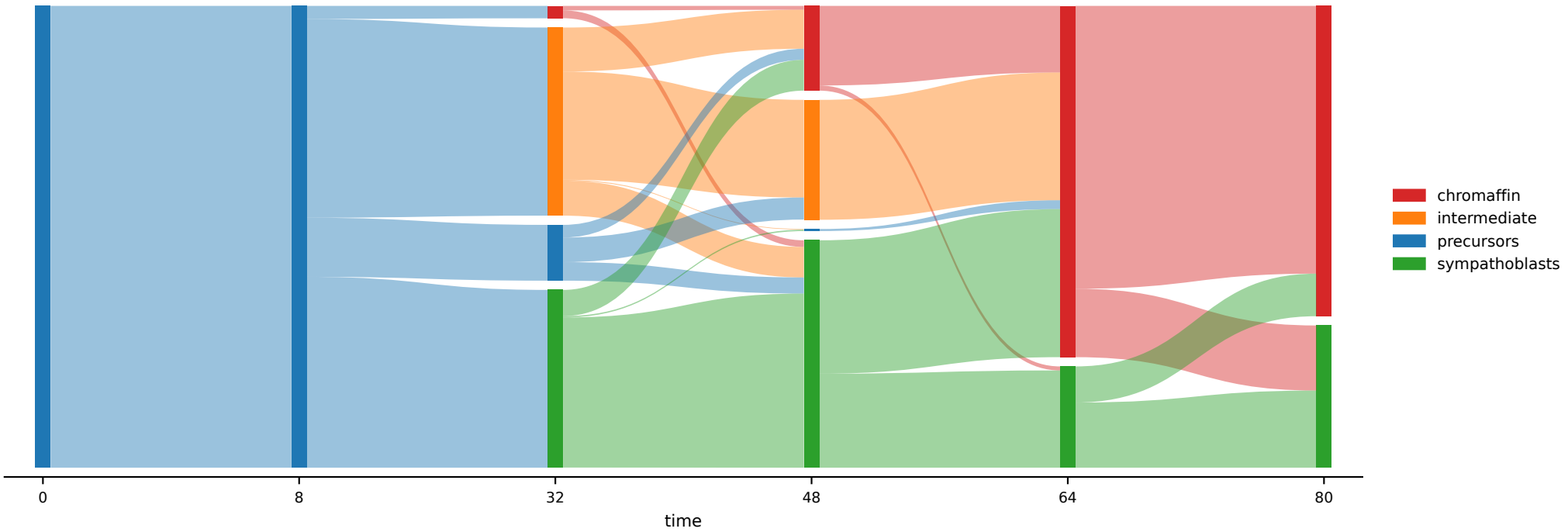
Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
chromaffin	**CHGA** **CHGB** MEG3 **DLK1** **PCSK1N** **TH** **SCG2** **RAMP1** **DBH** RGS4	**CHGA** **DLK1** **CHGB** **RAMP1** **TH** **SCG2** HTATSF1 **PCSK1N** PENK **DBH**
intermediate	**PLP1** PTPRZ1 **ZEB2** **POSTN** **COL5A2** QKI EDNRB SPARC VIM MPZ	MT-CYB **POSTN** **ZEB2** COL1A1 **COL5A2** VCAN ALDH1A1 **PLP1** OLFML2A NRXN1
precursors	**S100B** **VIM** **METRN** S100A10 **PLP1** **SPARC** ANXA2 **EDNRB** **MPZ** LGALS1	**PLP1** **S100B** **VIM** **SPARC** **EDNRB** PTPRZ1 **MPZ** **METRN** COL5A2 IFITM3
sympathoblasts	PTMA **GAP43** **TMSB10** **STMN2** **RTN1** **MAP1B** **BASP1** **GAL** **UCHL1** MDK	**STMN2** **RTN1** **BASP1** **GAP43** **MAP1B** **GAL** PHOX2A **UCHL1** CD24 **TMSB10**

Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
chromaffin	**CHGB CHGA PCSK1N SCG2 DBH** RGS4 **RAMP1 TH MEG3** RGS5	**CHGA** DLK1 **CHGB TH MEG3 RAMP1 SCG2** HTATSF1 **DBH PCSK1N**
intermediate	**COL5A2 POSTN PTPRZ1 ZEB2** QKI **PLP1** EDNRB NRXN1 **OLFML2A VCAN**	COL1A1 **POSTN ZEB2 VCAN COL5A2** NEFL **OLFML2A** ALDH1A1 **PTPRZ1 PLP1**
precursors	**S100B VIM PLP1** METRN **SPARC** S100A10 ANXA2 **EDNRB MPZ** IFITM3	**PLP1** PTPRZ1 **SPARC EDNRB** ZEB2 **VIM** COL5A2 **S100B MPZ** QKI
sympathoblasts	**PTMA STMN2 MAP1B** HAND2 **CD24 BASP1 PHOX2A UCHL1 GAL** HAND2-AS1	**STMN2** RTN1 **MAP1B GAL BASP1** GAP43 **PHOX2A CD24 PTMA UCHL1**

Genes of the model (97)

Data/data_train.h5ad, alphabetical order.

ACTG1	EIF4A2	NEFL	RPL41
ALDH1A1	FAU	NPY	RPS13
ANXA2	FOS	NRXN1	RPS2
APOE	GAL	OLFML2A	RPS29
ATP5F1E	GAP43	PCSK1N	RPS4X
B2M	GAPDH	PENK	RTN1
BASP1	H2AFZ	PHOX2A	S100A10
BEX1	HAND2	PLP1	S100A6
BTF3	HAND2-AS1	PNMT	S100B
C4orf48	HINT1	POSTN	SCG2
CARTPT	HMGB2	PPIA	SPARC
CD24	HMGN2	PRPH	STMN2
CHGA	HNRNPA1	PTMA	TH
CHGB	HTATSF1	PTMS	TIMP3
CNTN1	IFITM3	PTN	TMSB10
COL18A1	IGFBP2	PTPRZ1	TPM1
COL1A1	ITM2B	QKI	UBC
COL1A2	LGALS1	RAMP1	UCHL1
COL5A2	MAP1B	RGS4	VCAN
CRYAB	MDK	RGS5	VIM
DBH	MEG3	RPL10	YBX1
DLK1	MEG8	RPL12	ZEB2
EDNRB	METRN	RPL30	
EEF1A2	MPZ	RPL32	
EEF2	MT-CYB	RPL34	

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) when the depth factor s_i is applied.

Depth

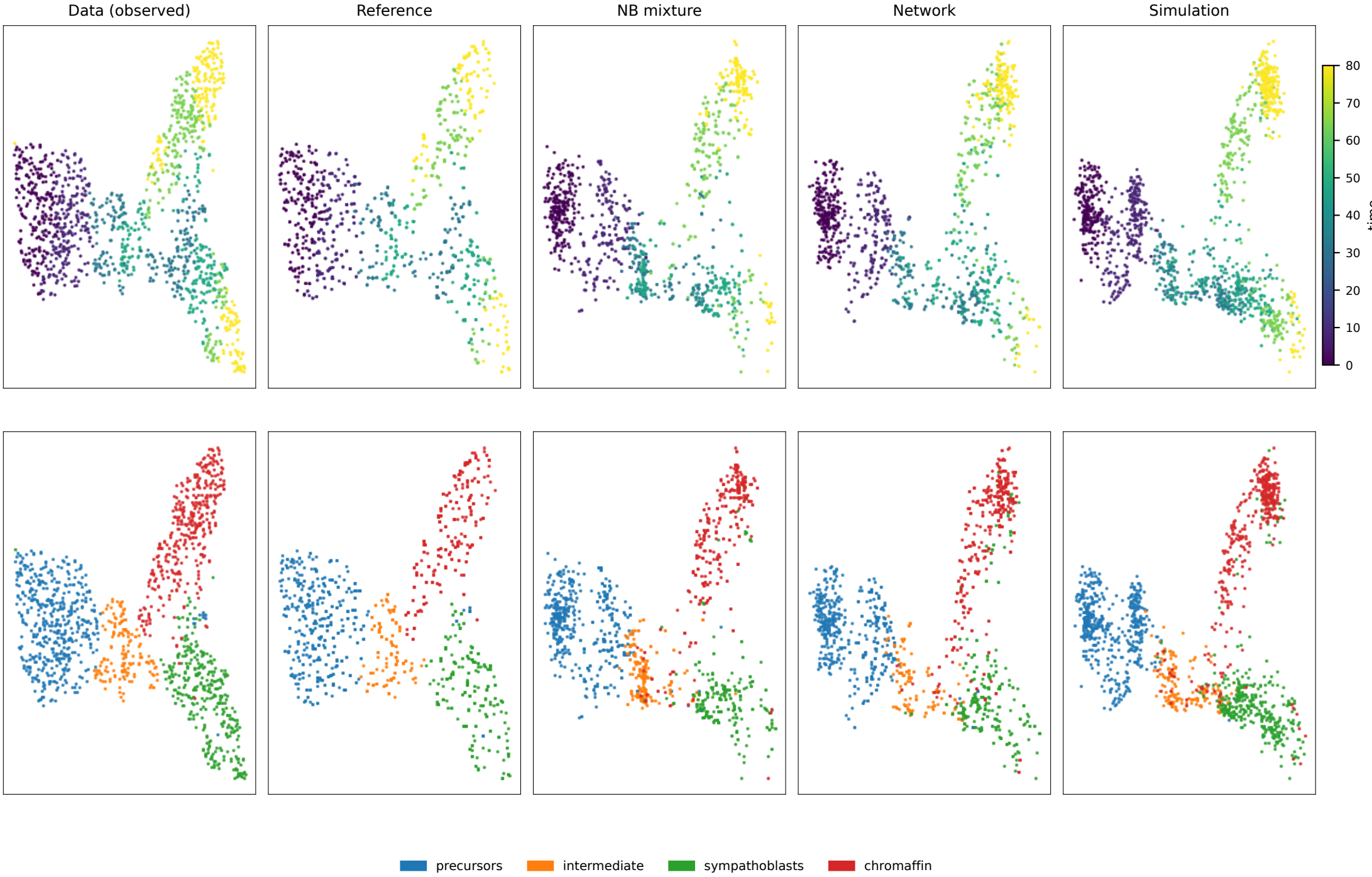
not estimated

Reason

97 genes < 10000

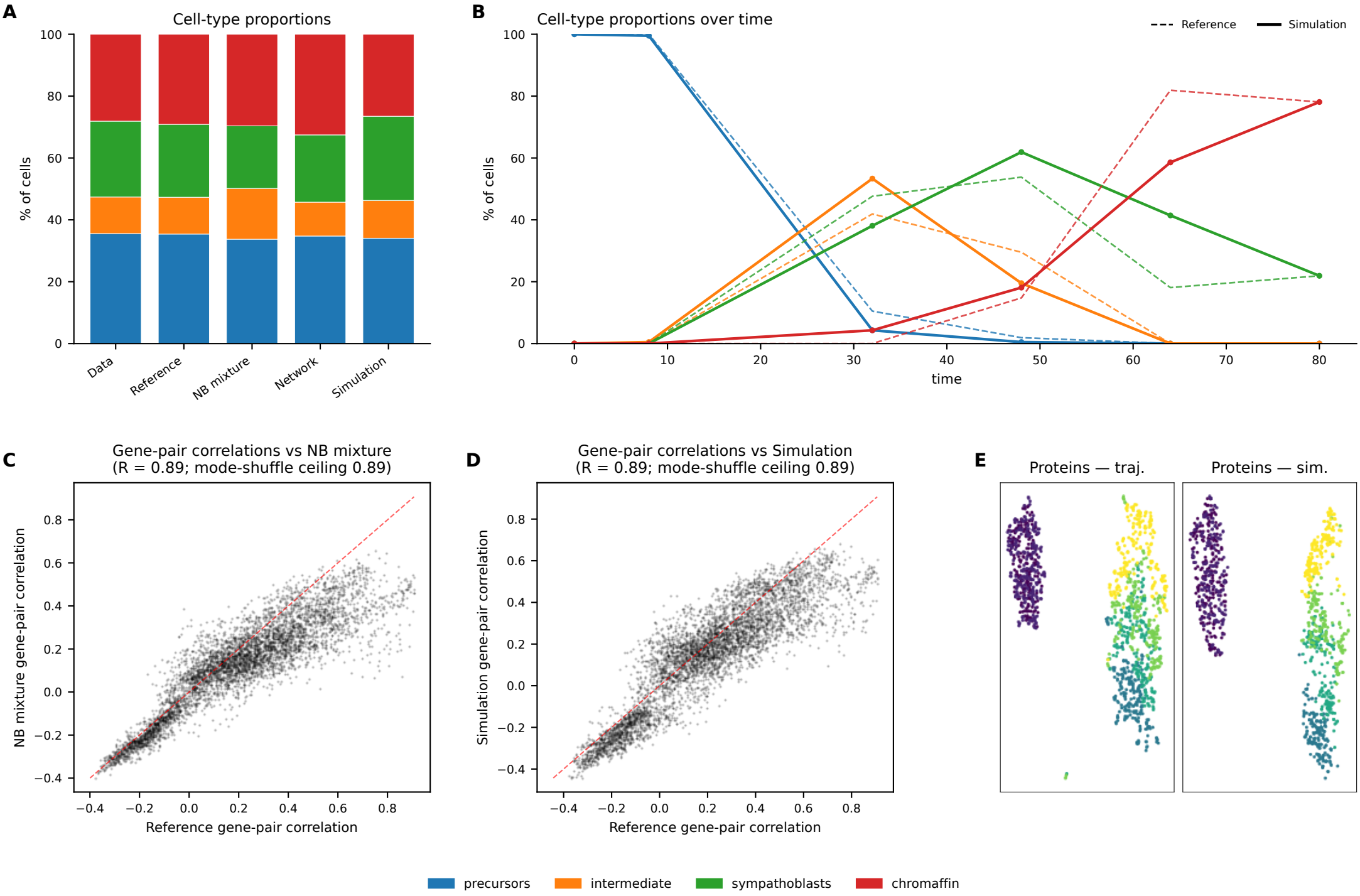
1. Generative model — trajectories and simulation

UMAP learned on the observed data; the trajectories (Reference: one state per ancestor, i.e. growth-weighted by exp ∫R_opt, as the simulation has proliferation), the NB mixture and network-driven modes along them and the full simulation are projected onto it.



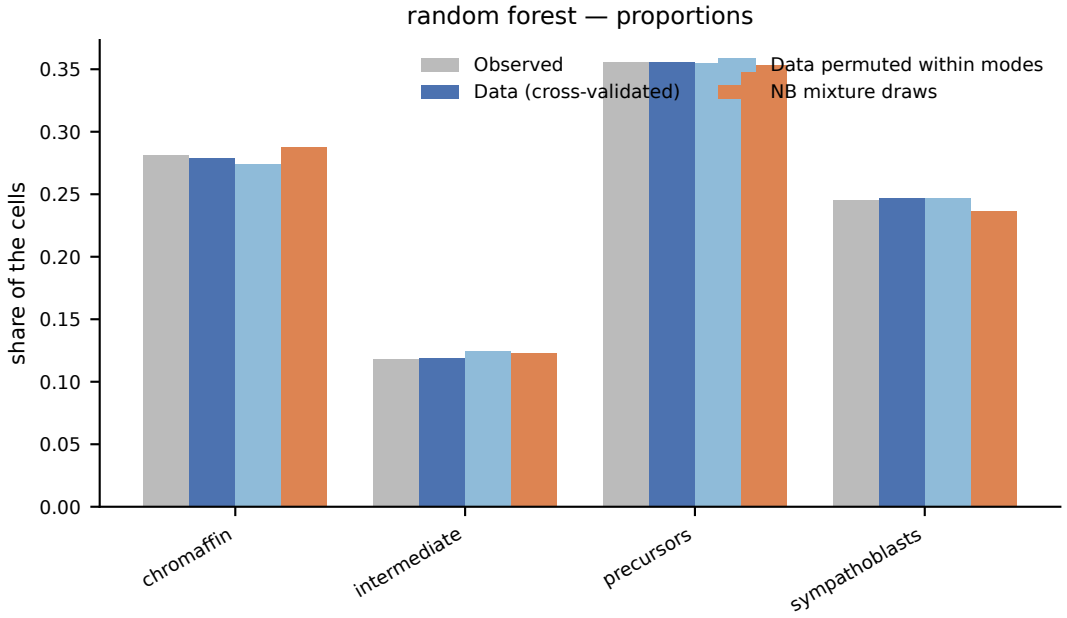
1. Generative model — quantitative checks

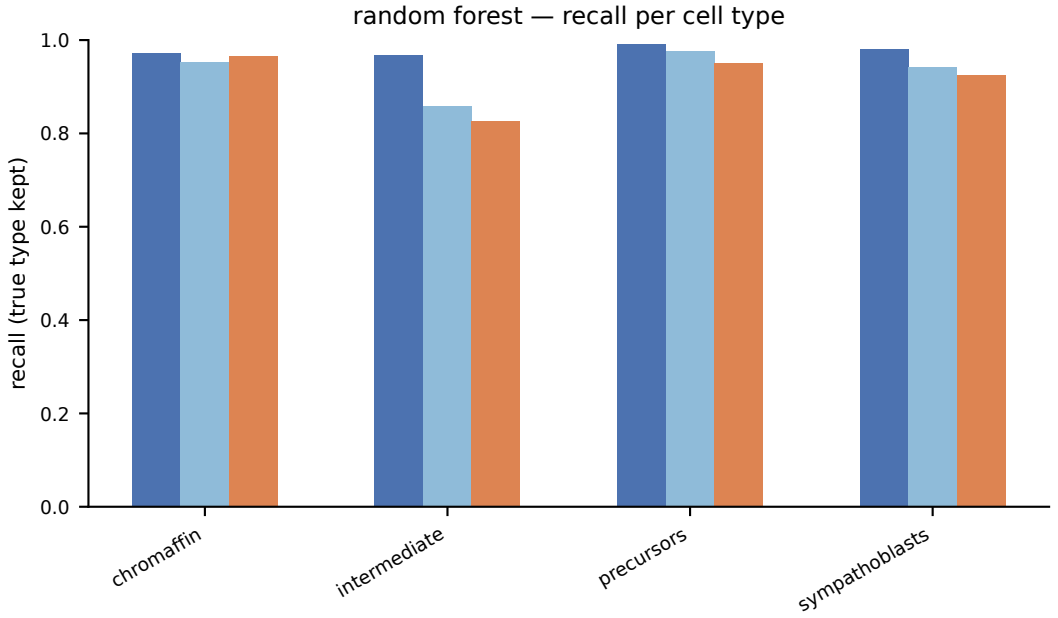
Reference = RNA of the trajectories equivalent to the simulation (growth-weighted: simulation with proliferation).

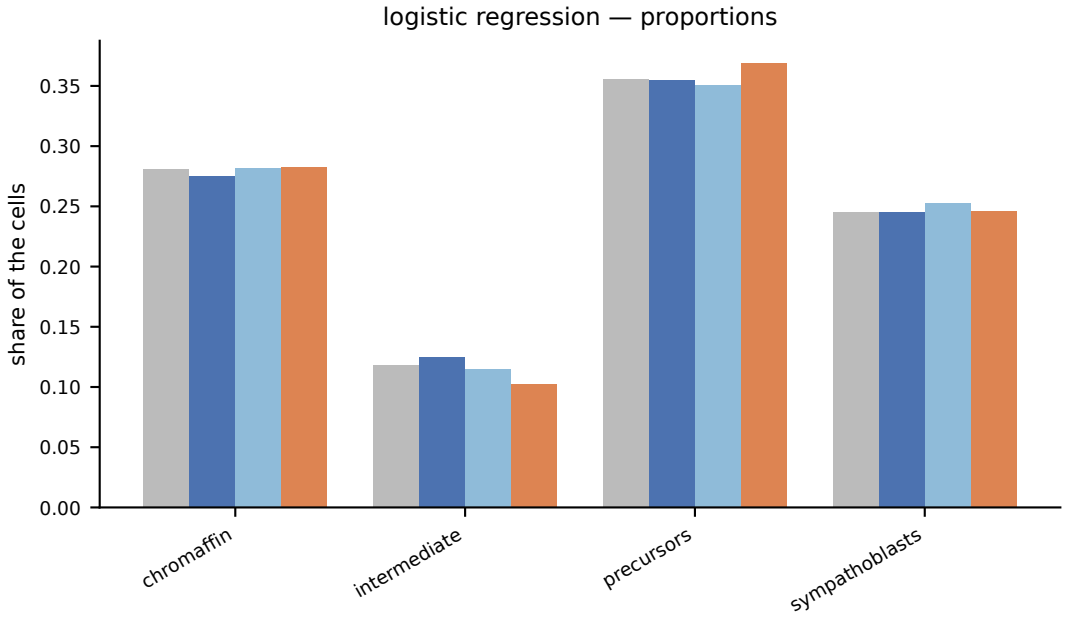


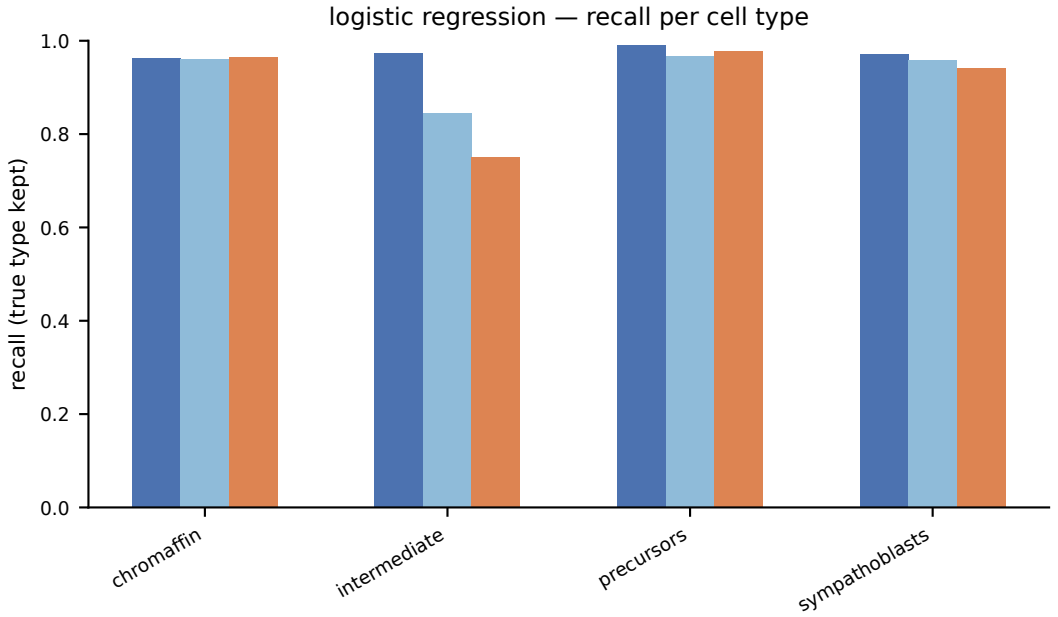
1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.





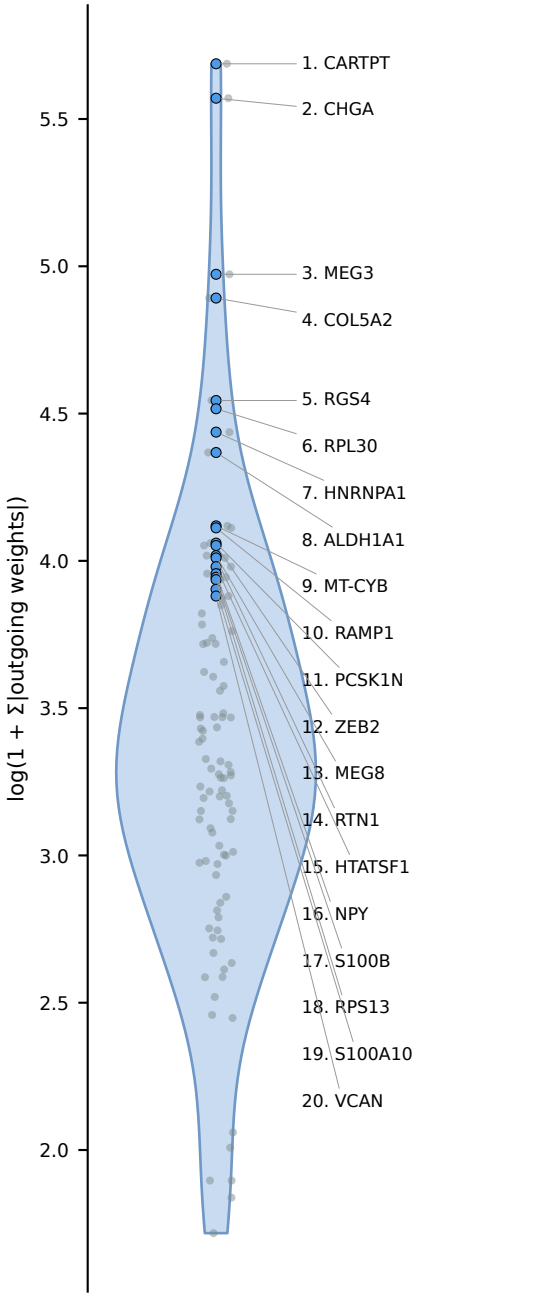




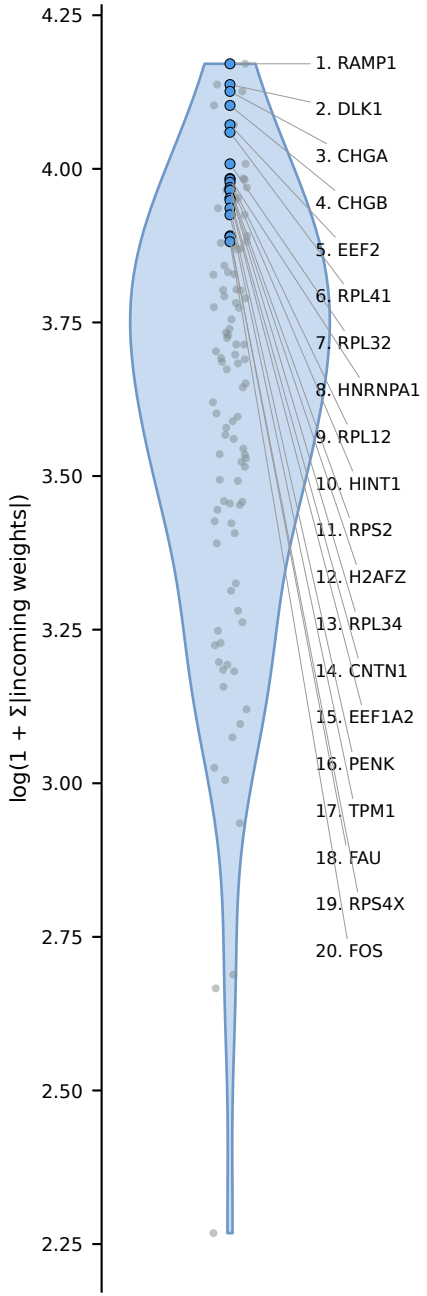
2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus not shown (stimulus = 0.2 < 0.5).

A



B



C Top 20 regulators

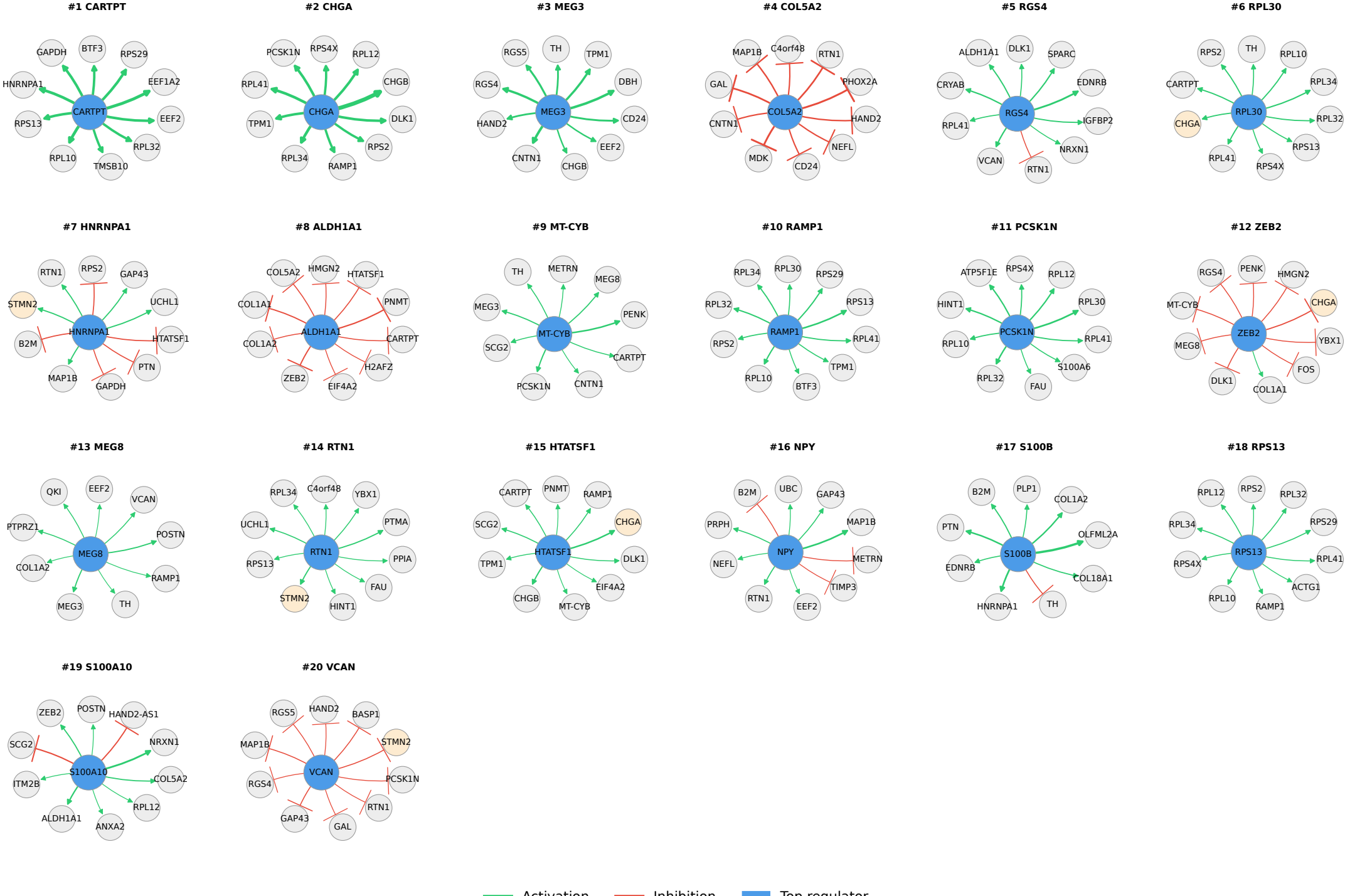
#	Regulator	Σ|w|	act.	inh.
1	CARTPT	294.13	60	1
2	CHGA	261.58	58	3
3	MEG3	143.43	46	9
4	COL5A2	132.23	9	43
5	RGS4	93.10	54	19
6	RPL30	90.50	54	17
7	HNRNPA1	83.53	17	41
8	ALDH1A1	77.87	5	65
9	MT-CYB	60.48	55	7
10	RAMP1	60.07	37	9
11	PCSK1N	56.99	34	0
12	ZEB2	56.52	3	49
13	MEG8	54.58	56	12
14	RTN1	54.15	42	11
15	HTATSF1	52.53	51	7
16	NPY	51.28	44	15
17	S100B	50.65	34	5
18	RPS13	50.22	43	8
19	S100A10	48.57	31	10
20	VCAN	47.45	6	54

D Top 20 regulated genes

#	Target	Σ|w|	act.	inh.
1	RAMP1	63.78	20	32
2	DLK1	61.62	22	23
3	CHGA	60.93	16	27
4	CHGB	59.53	18	23
5	EEF2	57.65	45	17
6	RPL41	56.95	32	17
7	RPL32	54.04	35	14
8	HNRNPA1	52.74	36	15
9	RPL12	52.66	41	13
10	HINT1	52.41	29	25
11	RPS2	51.96	30	16
12	H2AFZ	51.75	24	28
13	RPL34	51.73	42	9
14	CNTN1	51.06	21	18
15	EEF1A2	50.86	29	11
16	PENK	50.21	15	24
17	TPM1	49.66	31	19
18	FAU	47.95	29	15
19	RPS4X	47.88	25	10
20	FOS	47.49	29	25

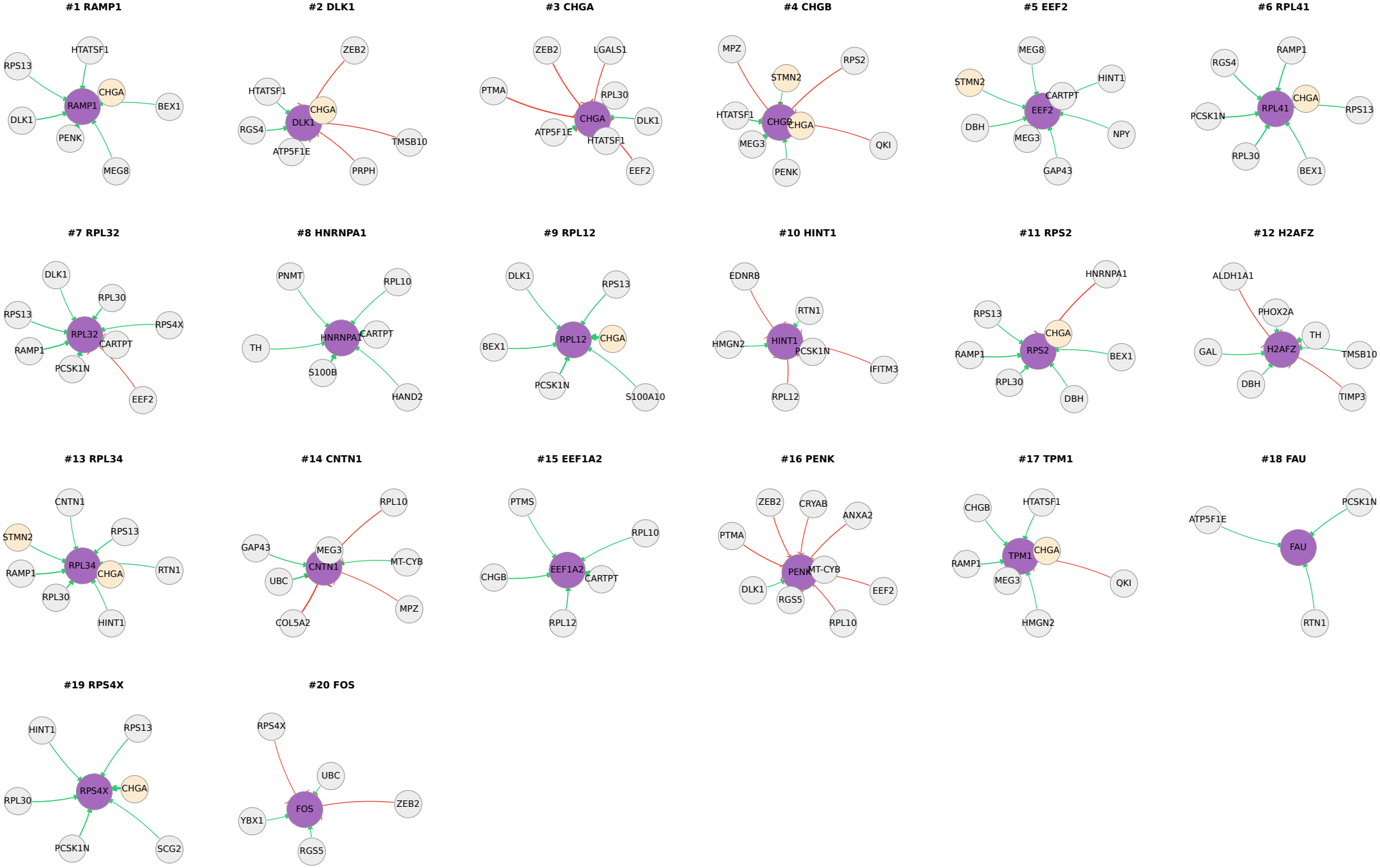
2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



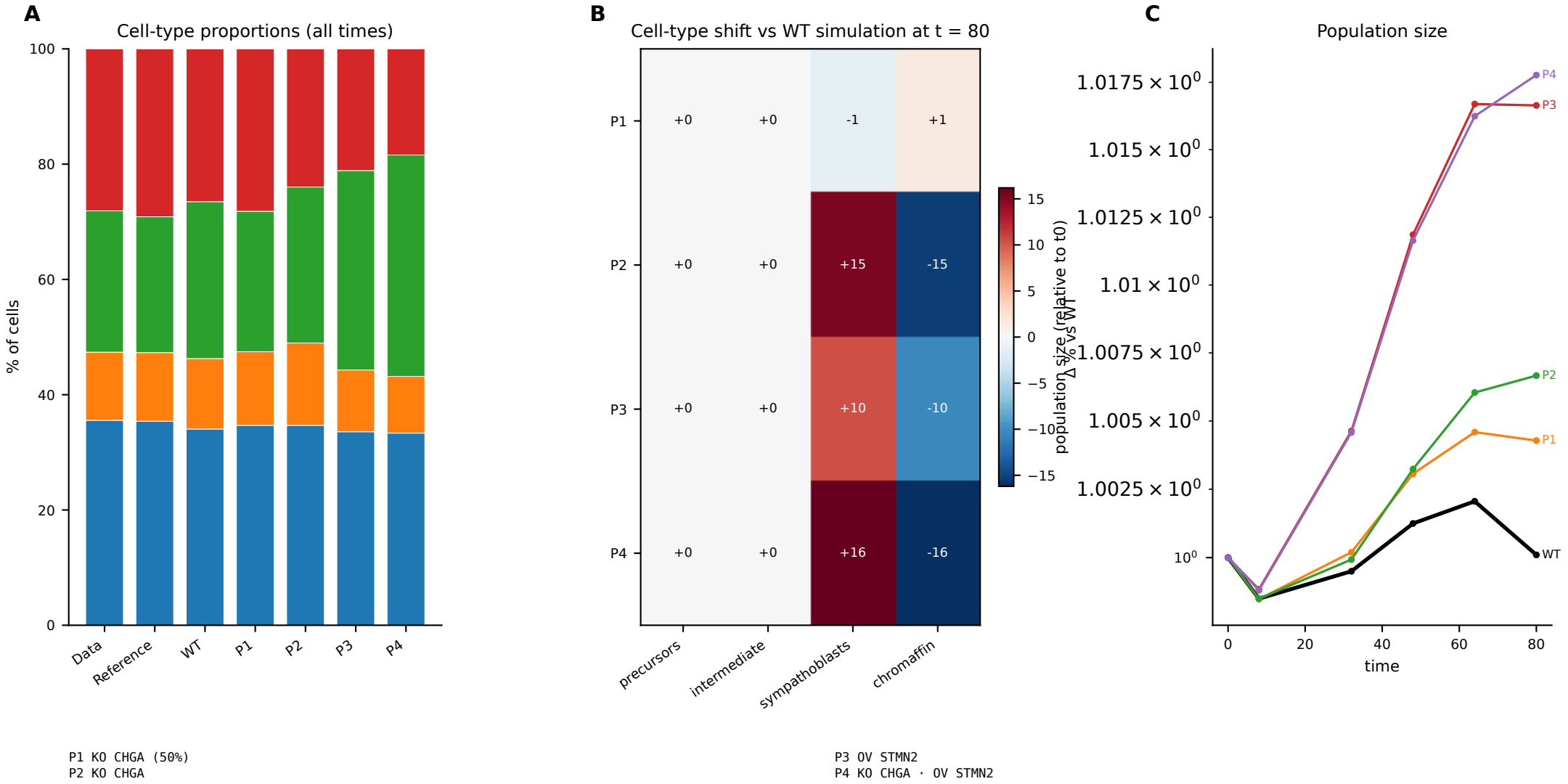
2. Gene regulatory network — top regulated genes and their main regulators

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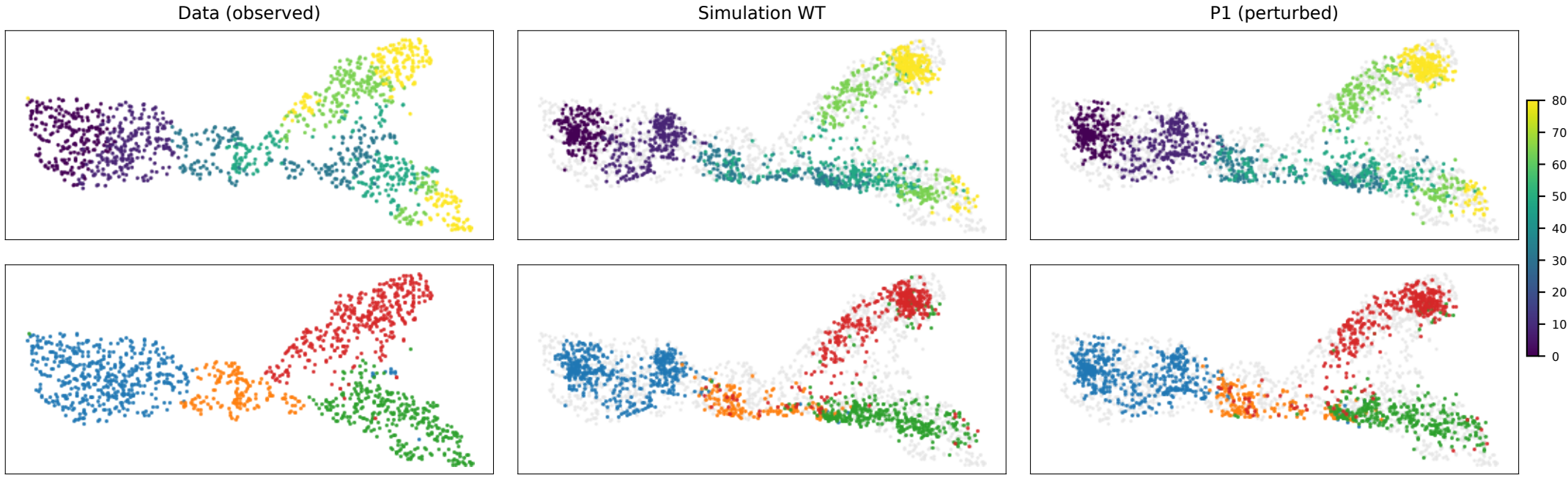
3. In-silico perturbations — overview

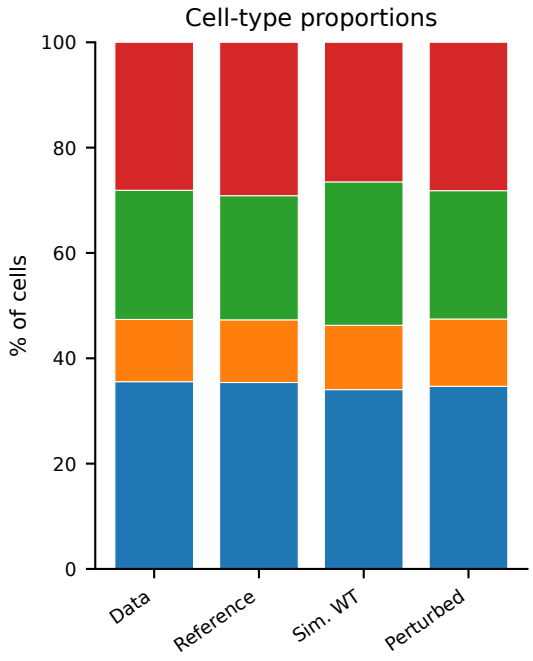
Cell types predicted by per-sample classifiers (logistic) trained on the observed data. Population sizes from the branching simulation (proliferation MLP + RATE effects).

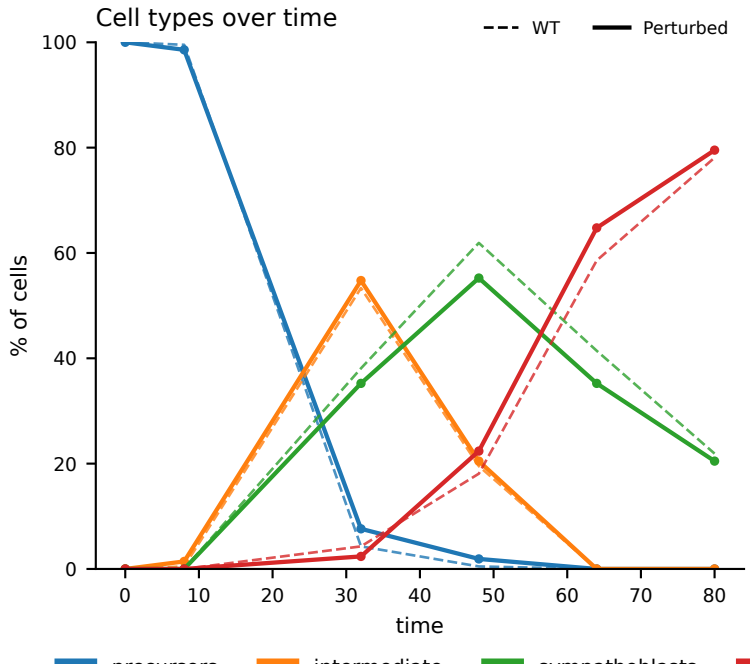


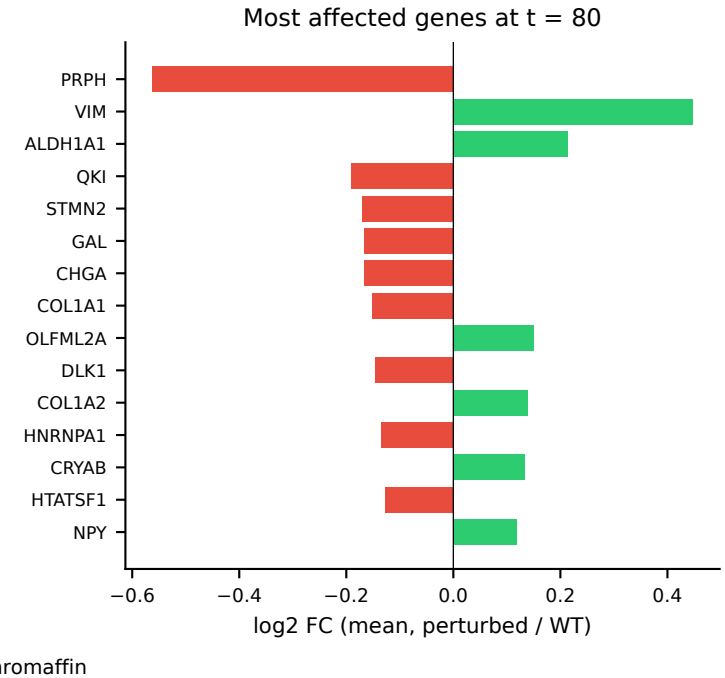
3. Perturbation P1 — KO CHGA (50%)

KO_CHGApct50_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 80: ×1.00 vs WT.



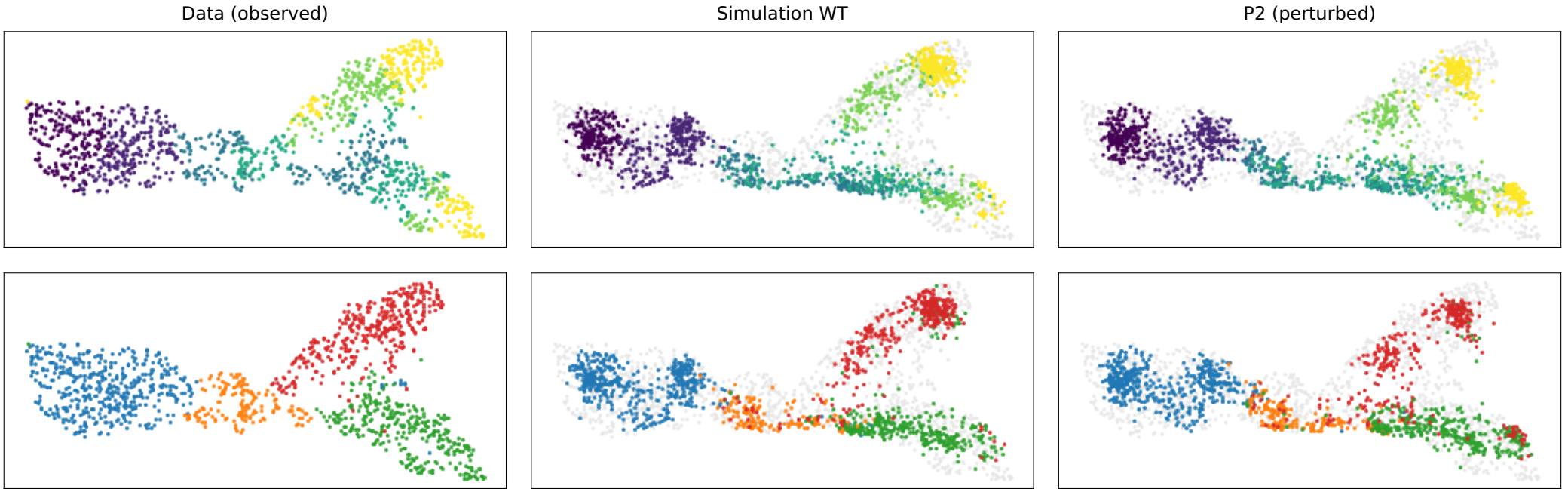


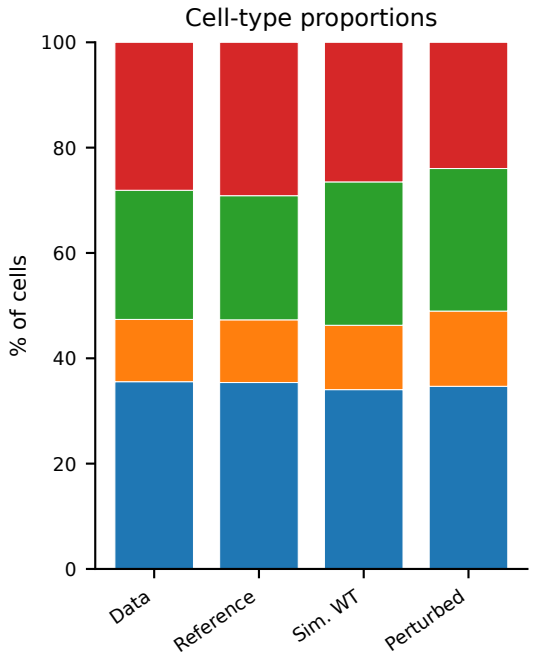


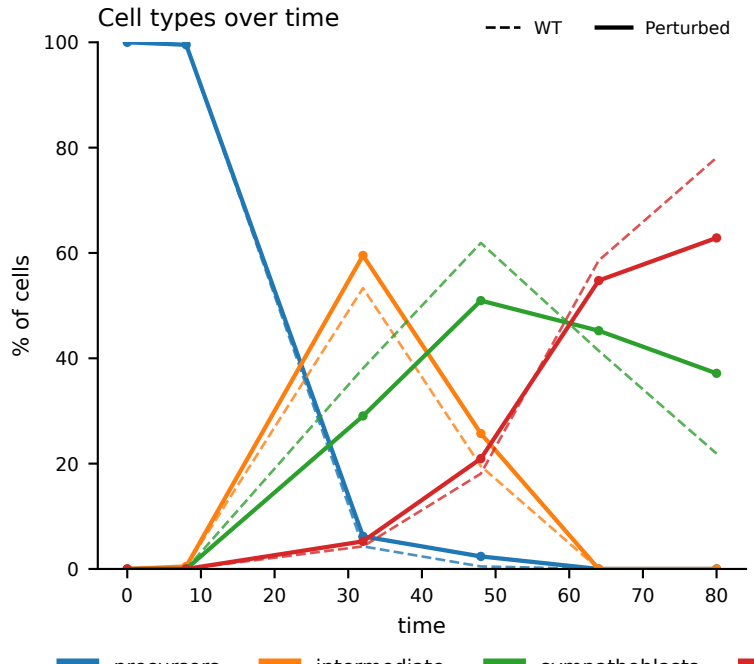


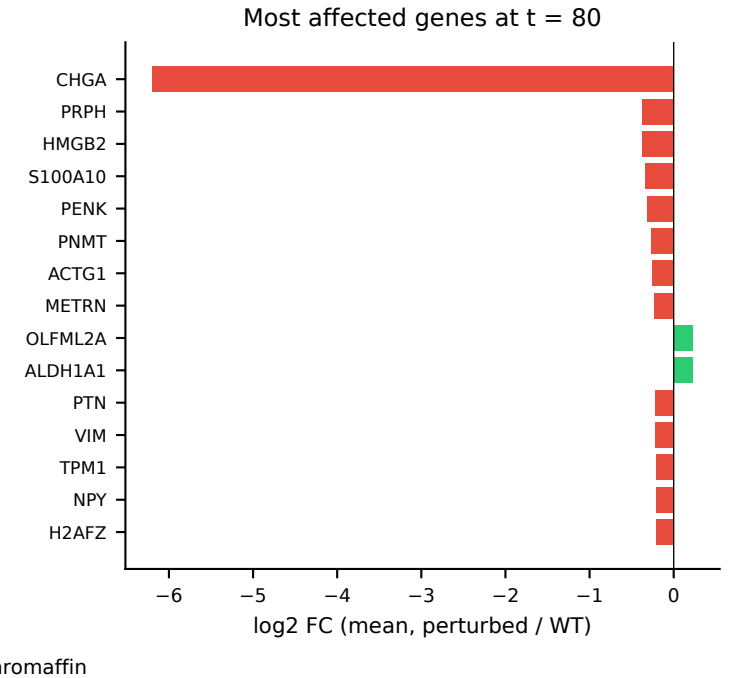
3. Perturbation P2 — KO CHGA

KO_CHGA_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 80: ×1.01 vs WT.



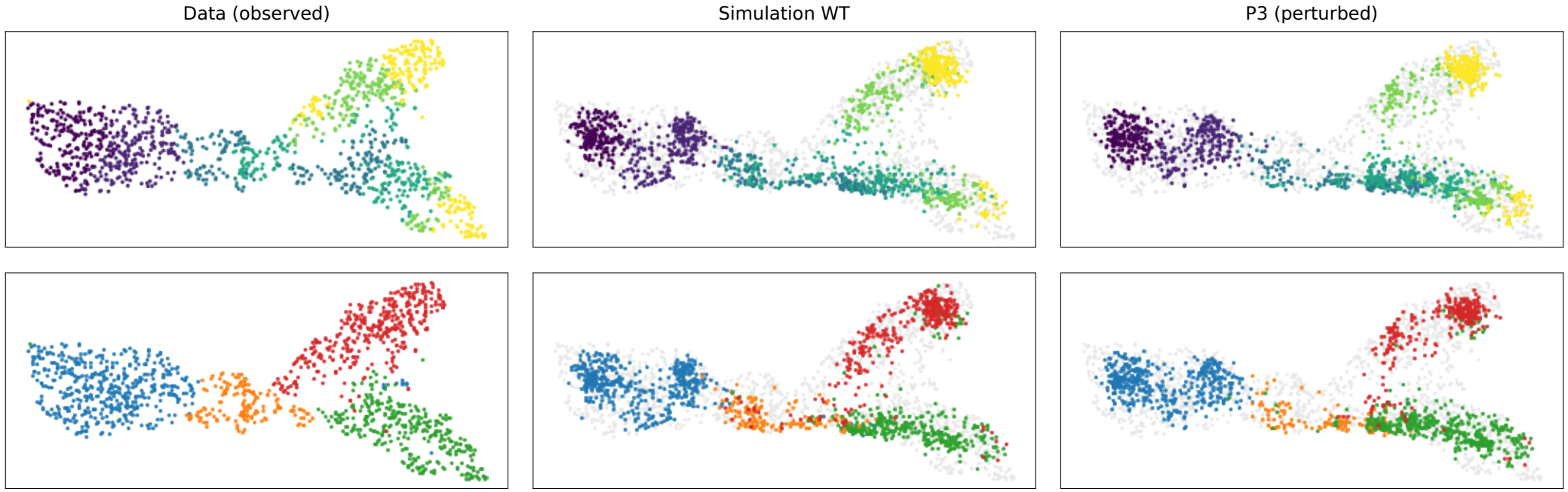


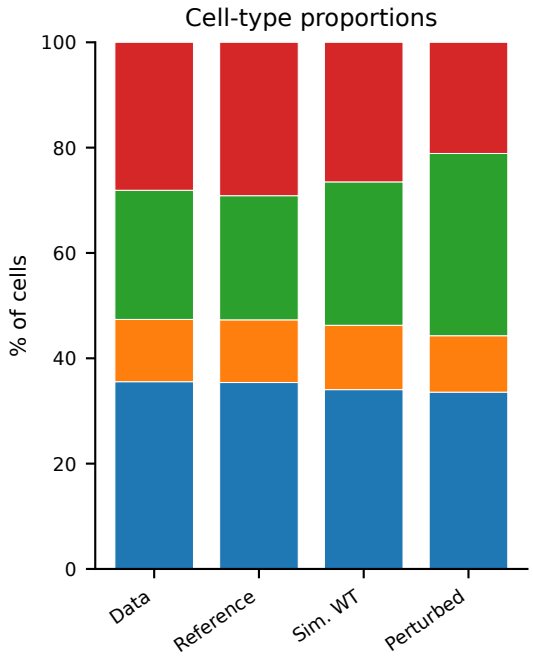


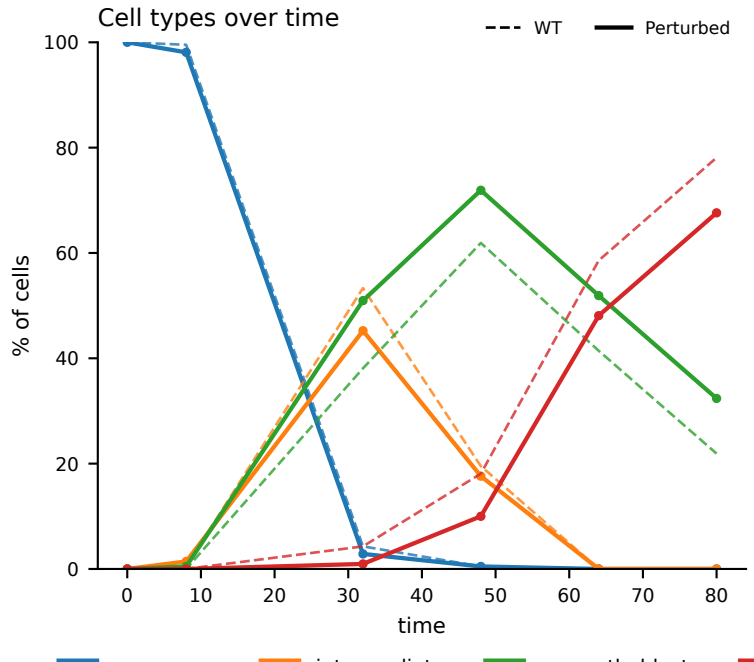


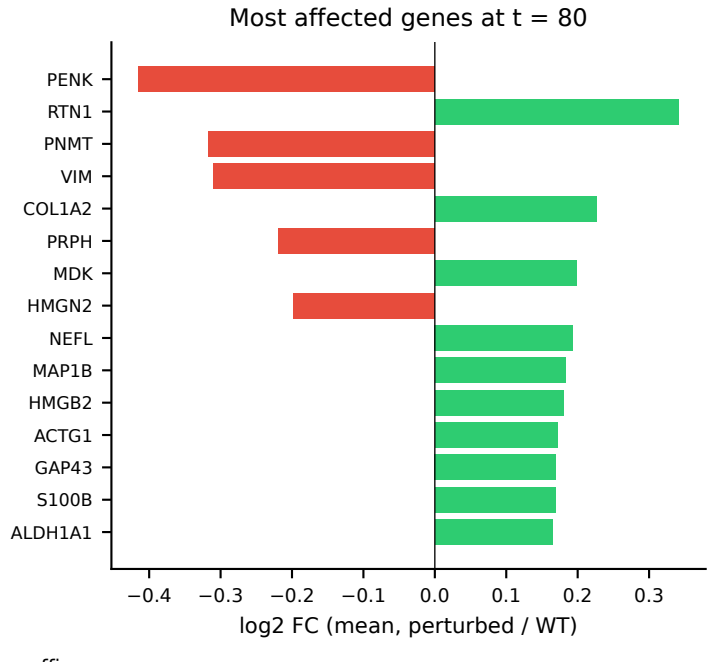
3. Perturbation P3 — OV STMN2

KO_none_OV_STMN2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 80: ×1.02 vs WT.



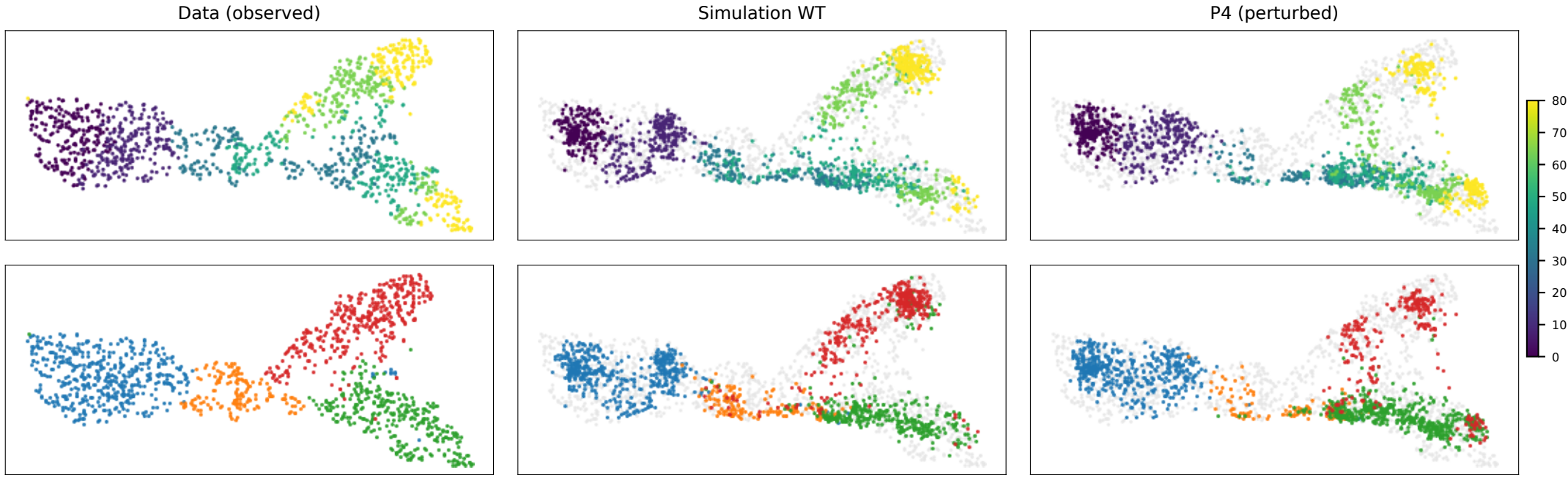


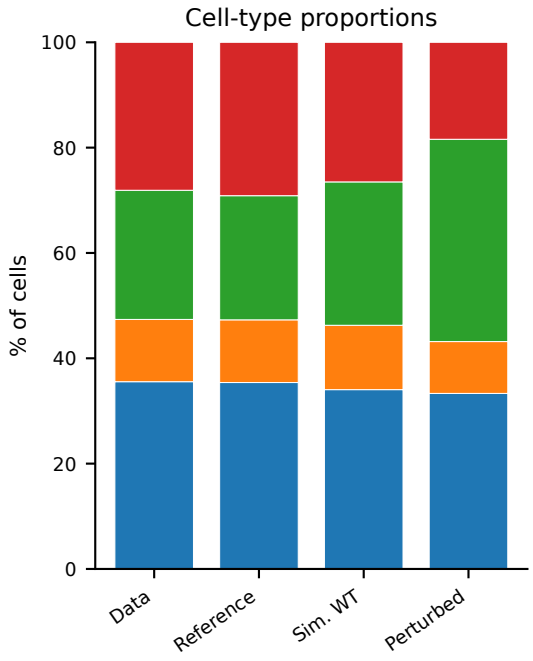


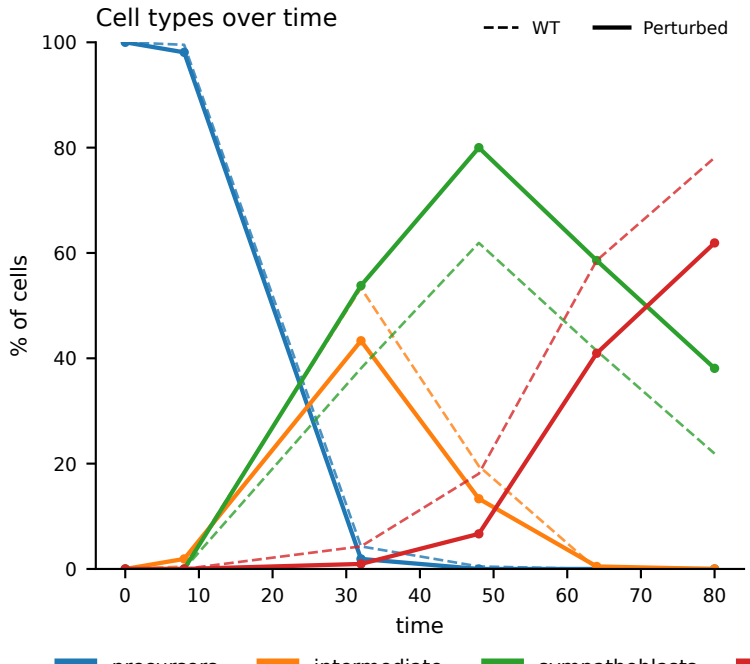


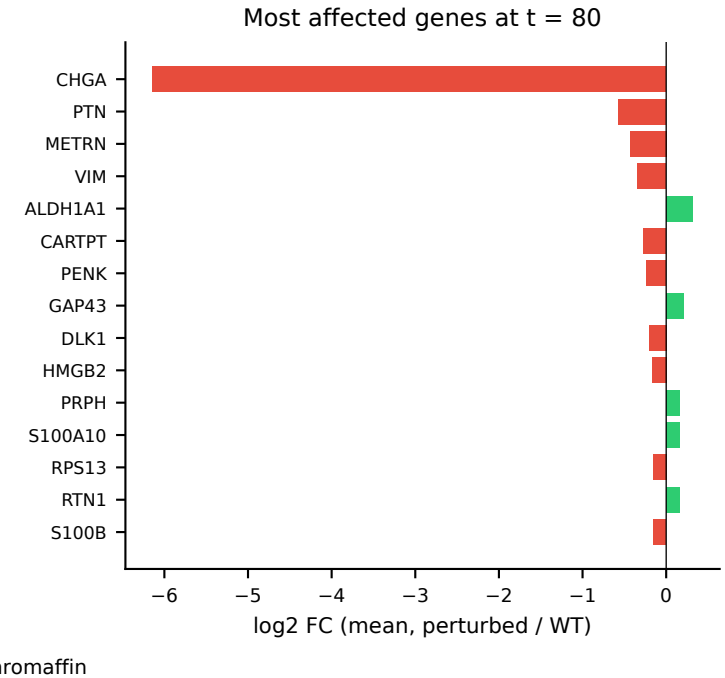
3. Perturbation P4 — KO CHGA · OV STMN2

KO_CHGA_OV_STMN2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 80: ×1.02 vs WT.







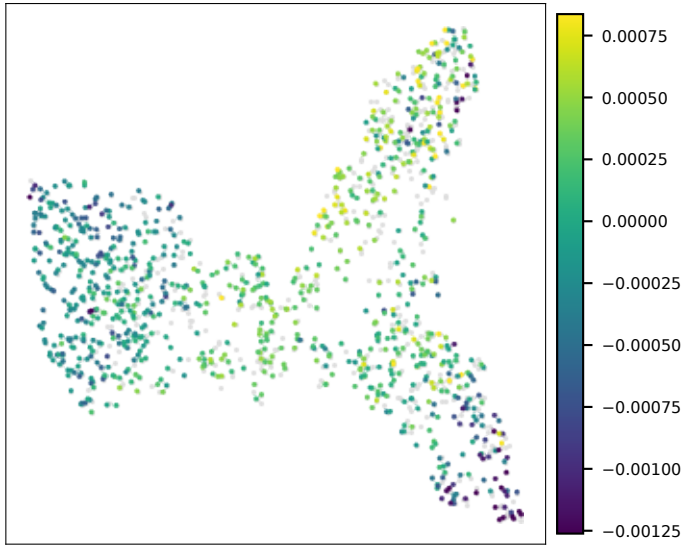


4. Proliferation — prior vs learned net rate (birth − death)

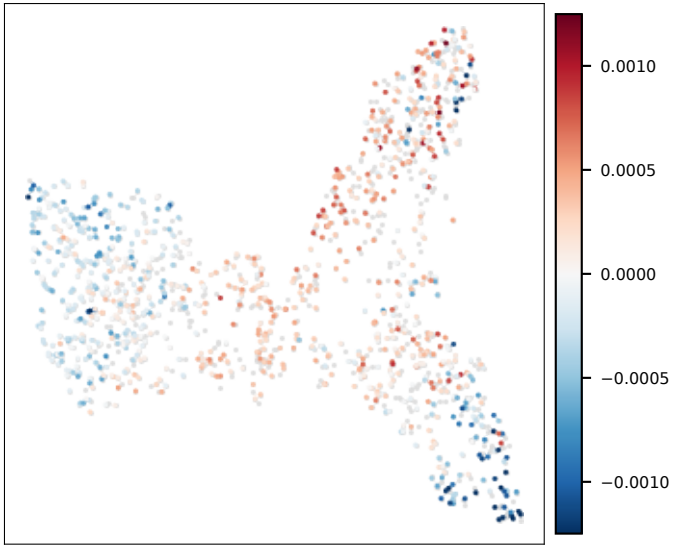
Learned = MLP R(P), mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.

No prior provided:
uniform net rate
(only relative rates are learned)

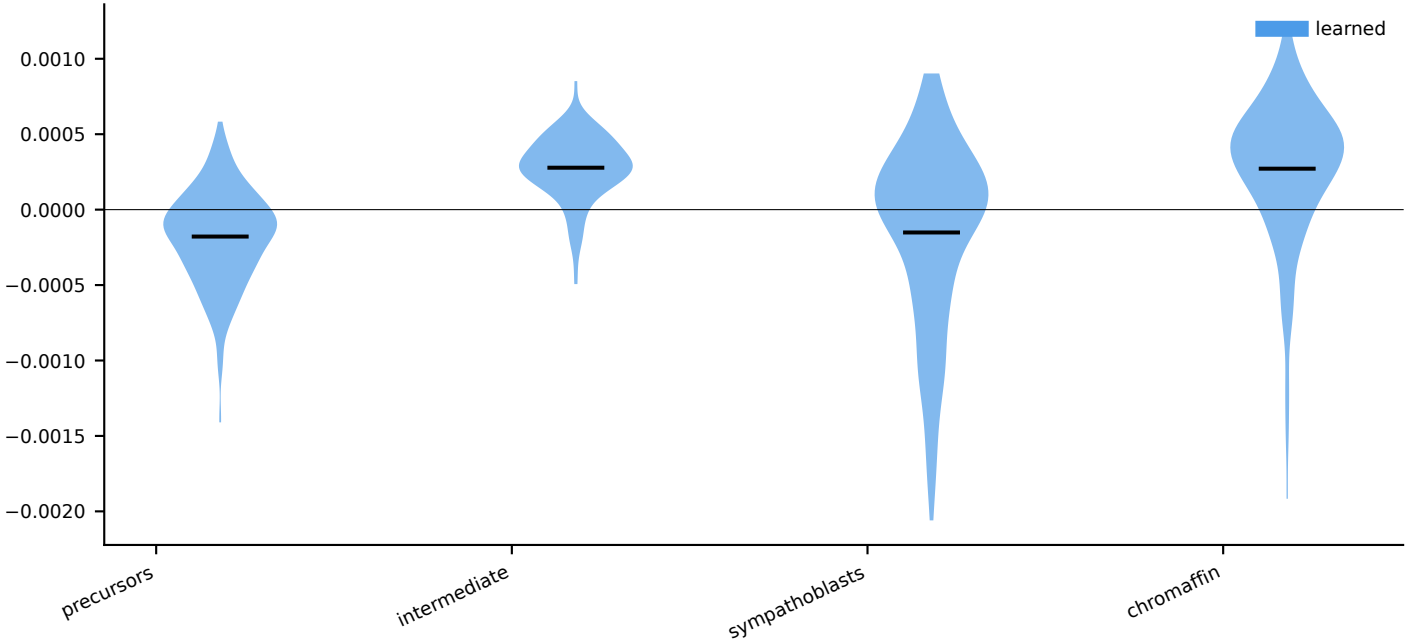
Learned net rate (MLP R(P))



Learned − mean



Net proliferation rate per cell type (cell_type)

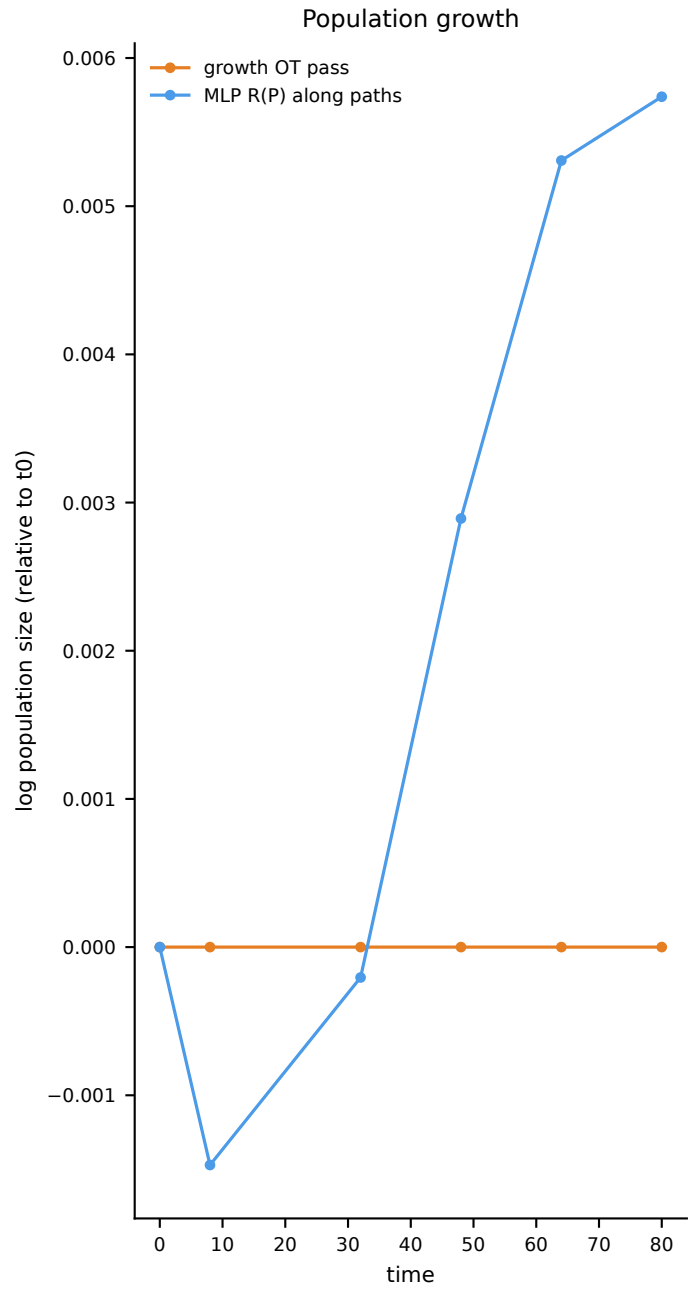


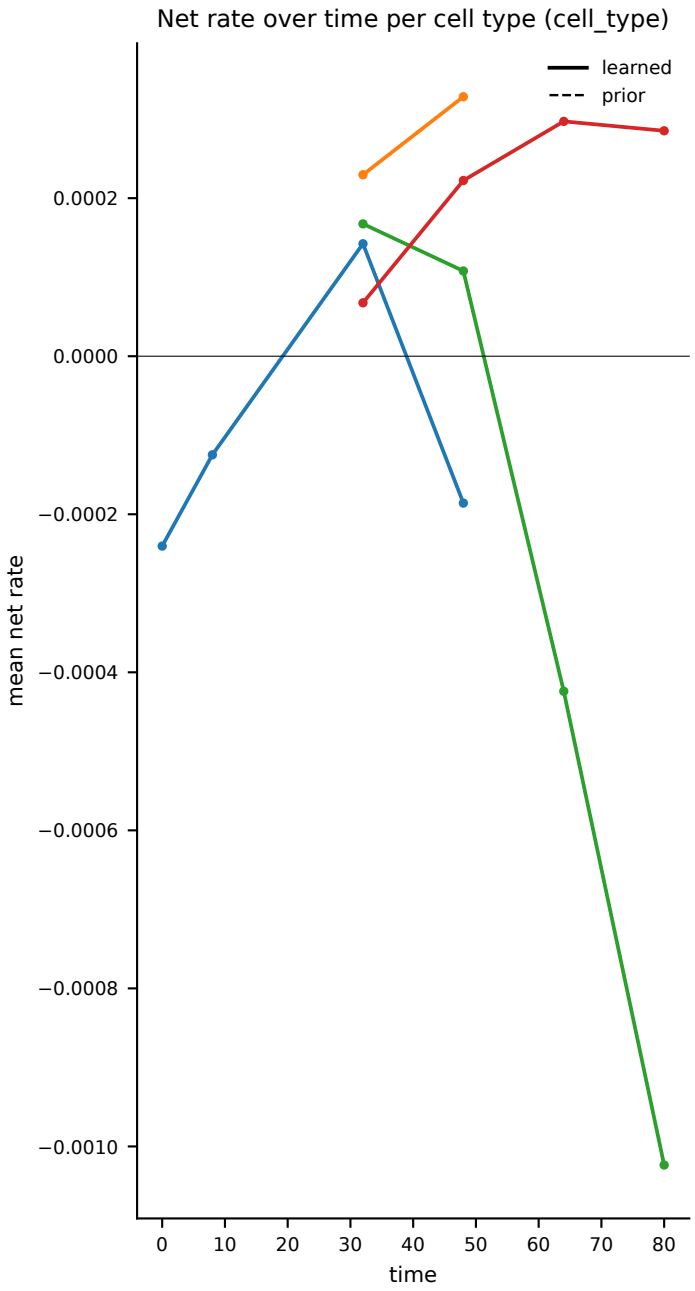
Mean net rate per cell type (cell_type)

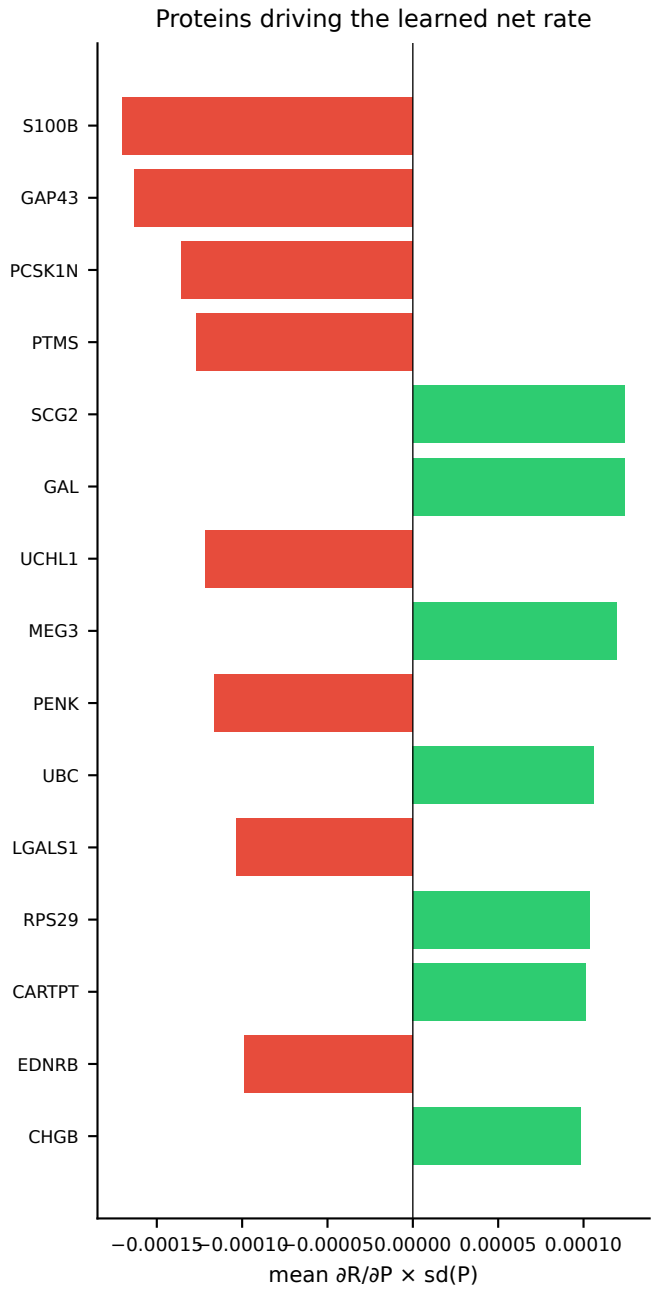
Cell type	Anchor	Prior	Learned
precursors	—	—	-0.000178
intermediate	—	—	0.000278
sympathoblasts	—	—	-0.000151
chromaffin	—	—	0.000271

4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; its absolute level follows the prior (anchored on the sheet population_sizes by fit_population_anchors, if filled).



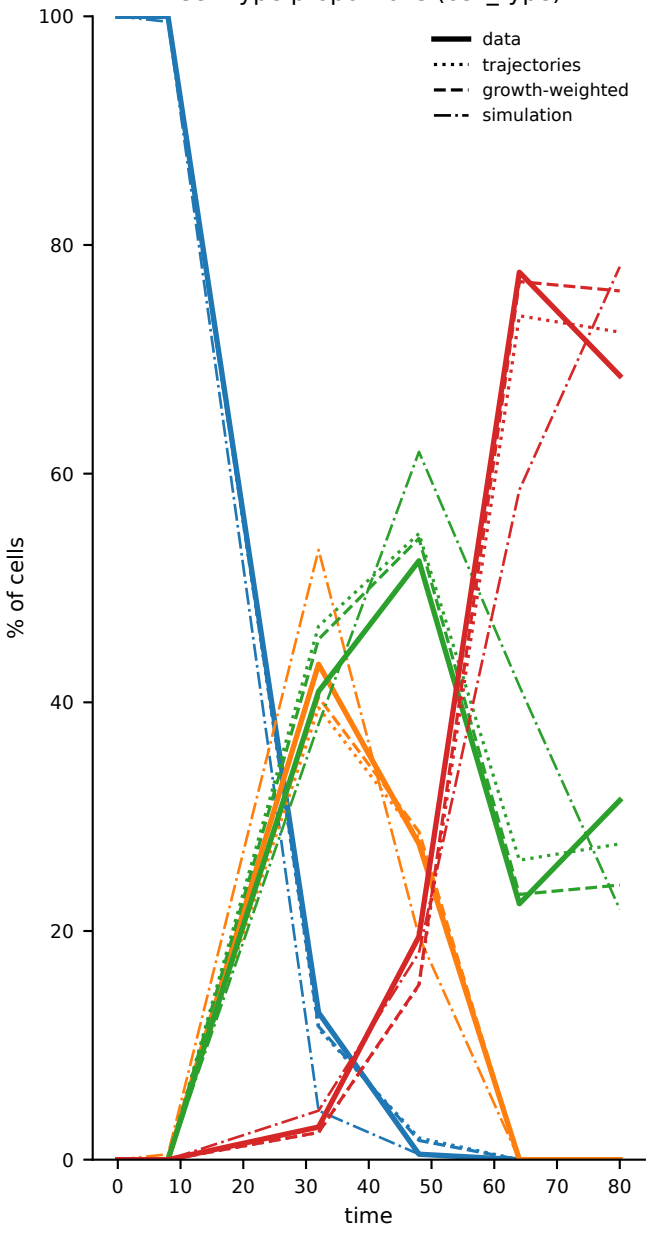


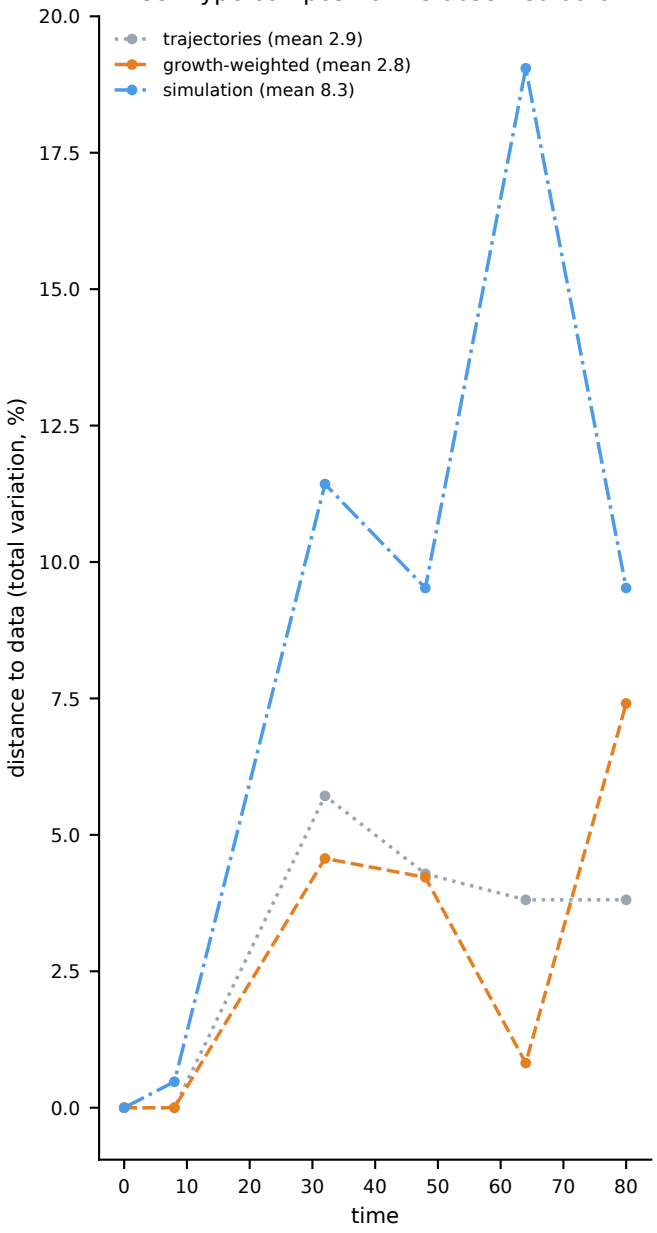


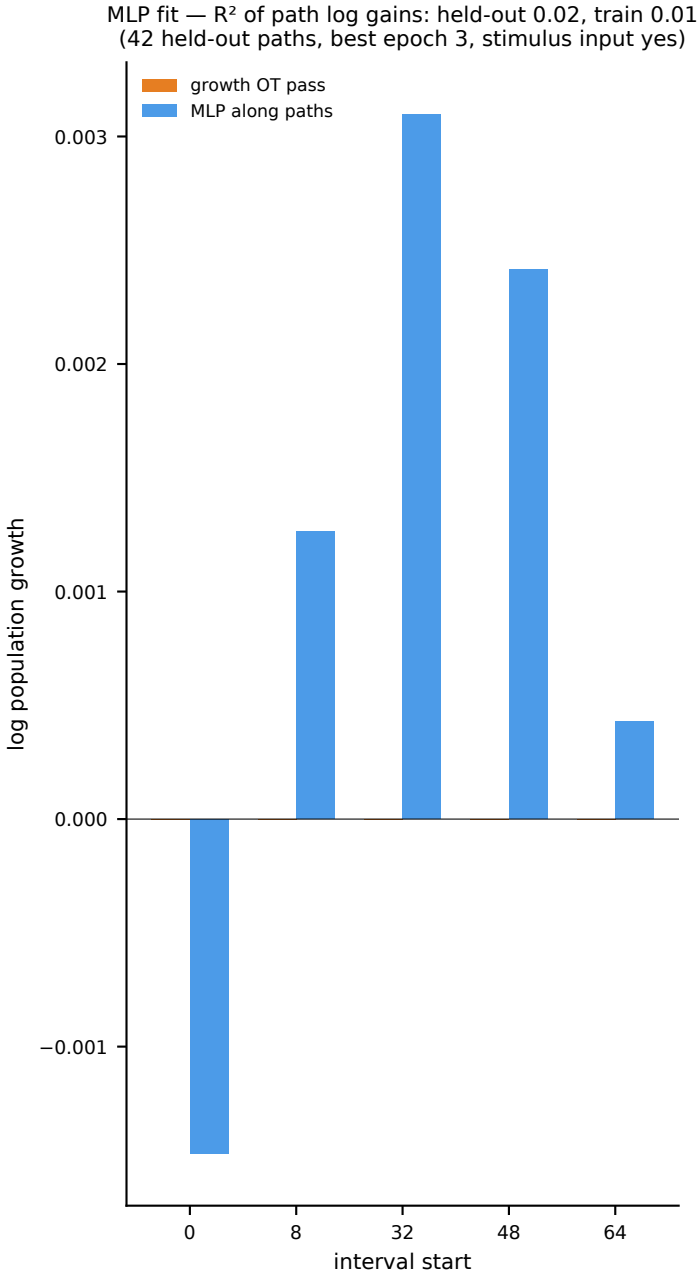
precursors intermediate sympathoblasts chromaffin

4. Proliferation — growth-weighted trajectories and MLP fit

The OT selection keeps one descendant per ancestor (trajectories without proliferation); weighting each path by its expansion exp(∫R_opt) should bring them back to the observed data. The simulation used the proliferation MLP.









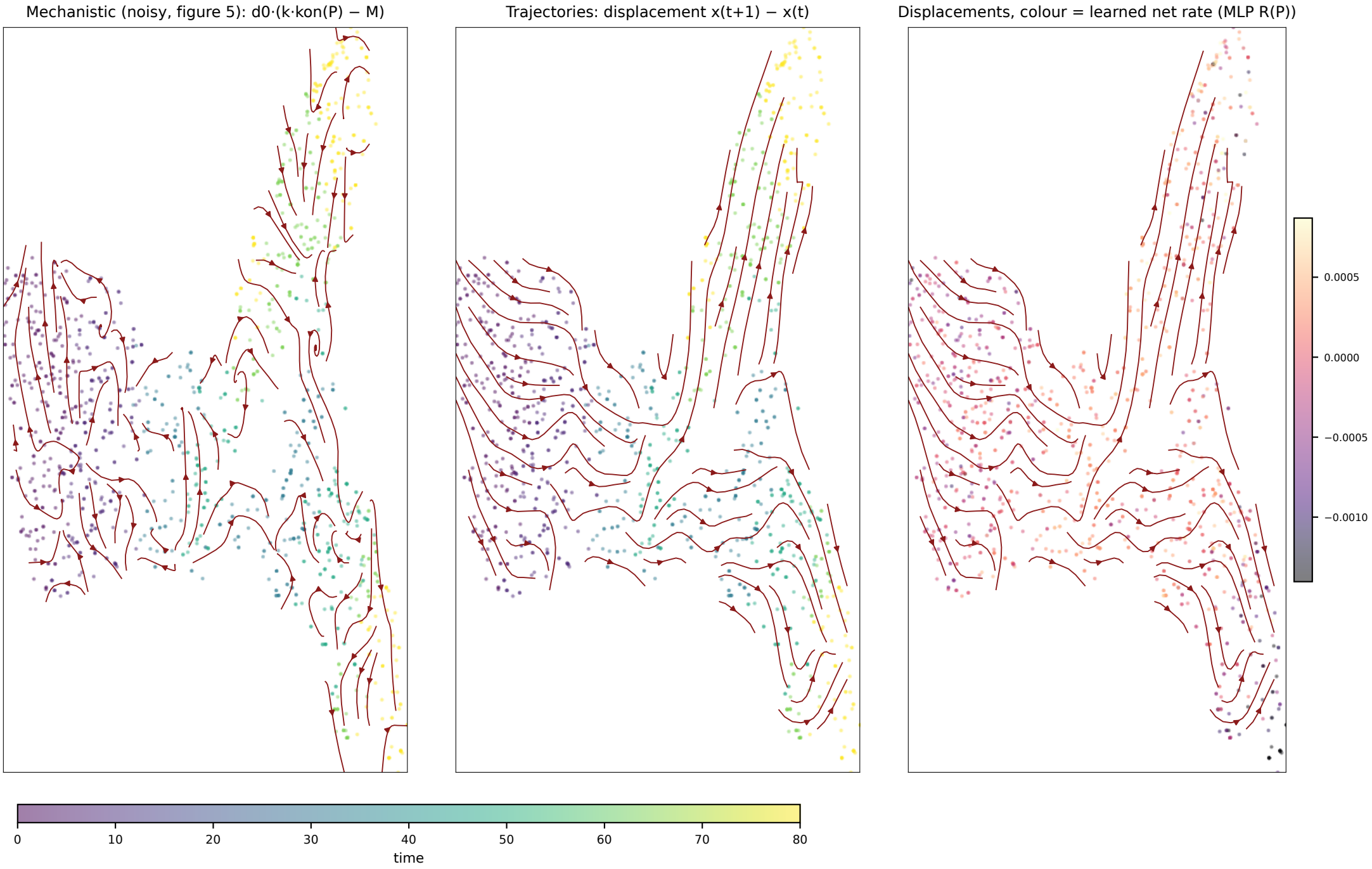
4. Protein dilution

Section unavailable:

```
ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
```

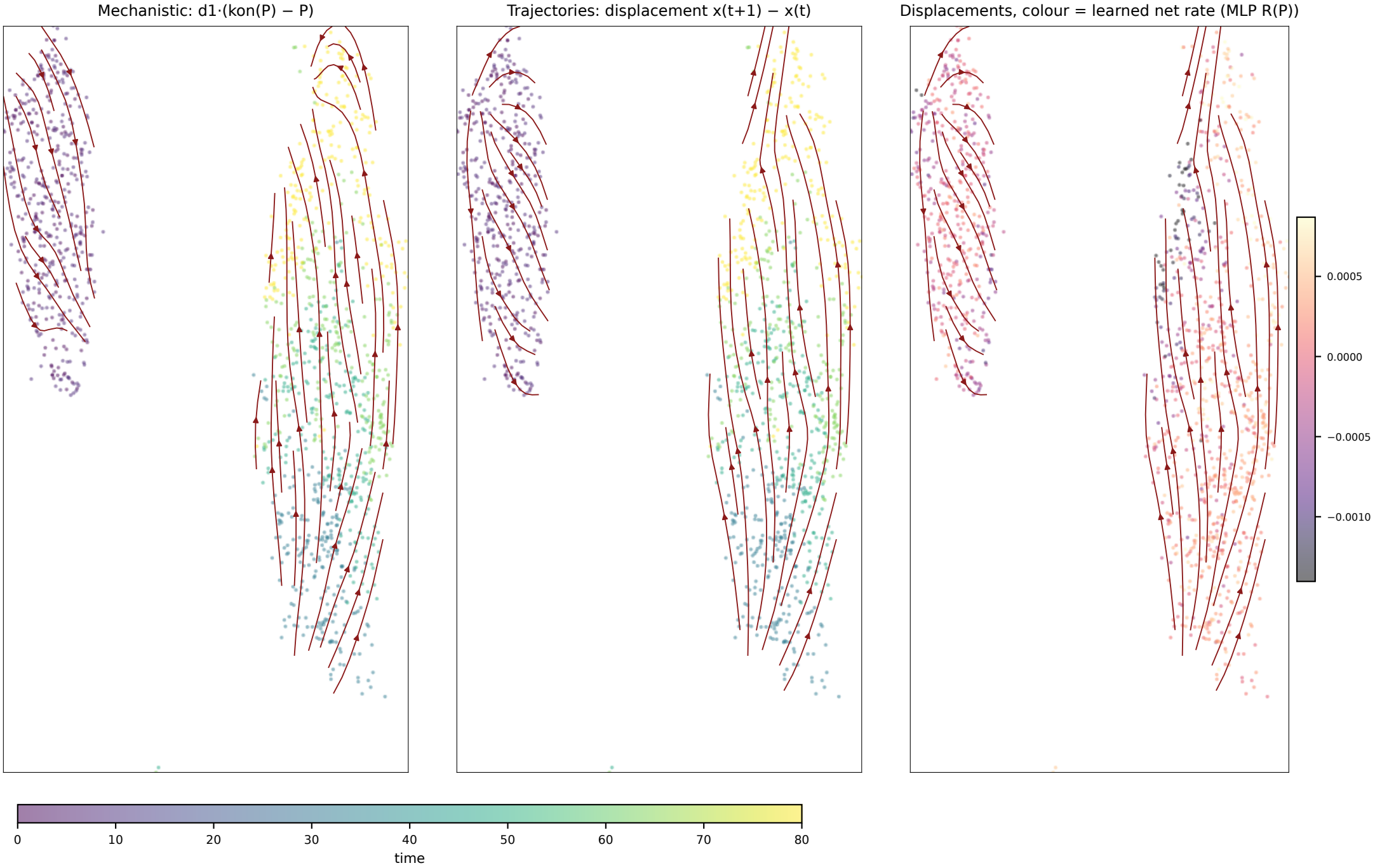
5. Learned dynamics — mRNA velocity and displacement fields

All 1260 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.05.



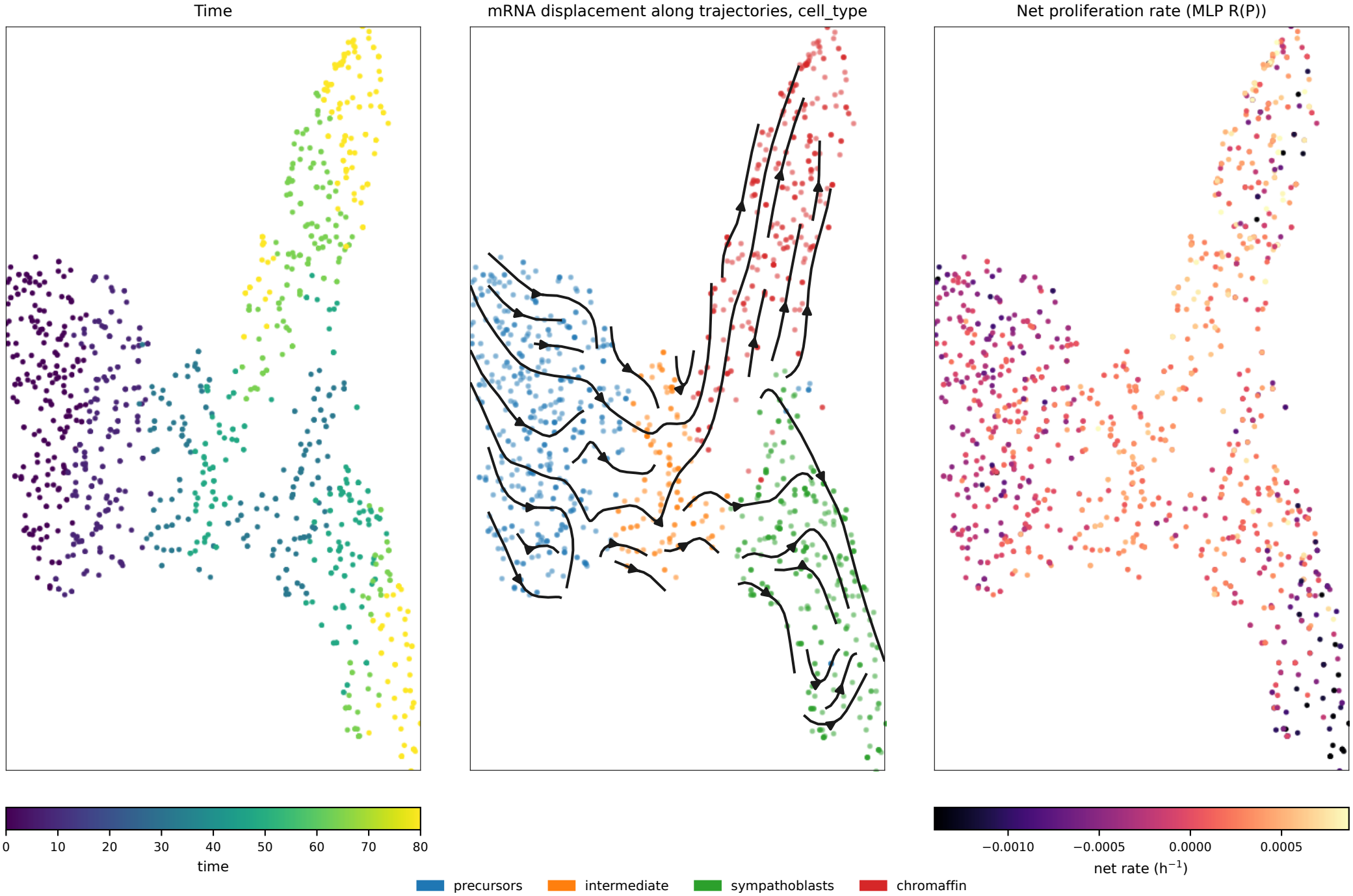
5. Learned dynamics — Proteins velocity and displacement fields

All 1260 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.54.



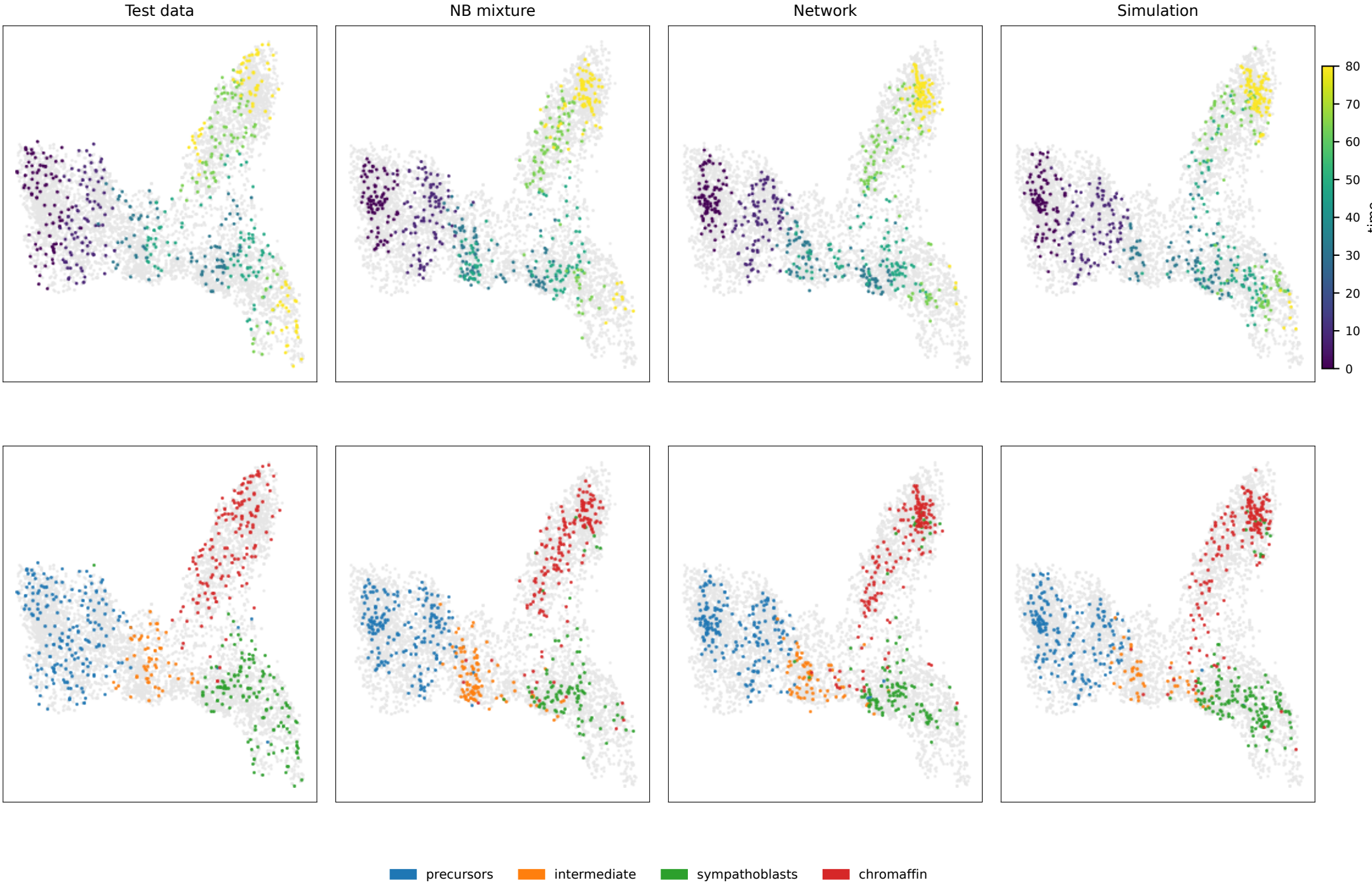
5. Learned dynamics — summary

All 1260 trajectory states on the UMAP learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.



6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

